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Surgical Sagittal Blade Cartridge

Abstract

A surgical sagittal saw cartridge that includes a guide bar formed from an inner plate and opposed outer plates. The cartridge has a blade that extends distally from the guide bar, the blade comprising a blade head including a plurality of teeth. The plurality of teeth may include a combination of pilot teeth and base teeth, the pilot teeth configured to extend distally beyond the base teeth. The inner plate may comprise an inner tine and/or two opposed outer tines. The inner tine may be formed to define the head against which the blade is configured to pivot about. The inner plate may be configured to have opposed edges that extend beyond the opposed edges of the outer plates to define a mounting feature. The inner plate may also define a reference feature configured to register and/or verify the pose of the cartridge with a navigation system.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS [0001] This application claims the benefit of priority to U.S. Provisional Application No. 63/127,727, filed Dec. 18, 2020, and the benefit of priority to U.S. Provisional Application No. 63/251,898, filed Oct. 4, 2021, the disclosures of both of which are incorporated herein by reference in their entirety.

BACKGROUND

[0002] A sagittal saw blade is a surgical saw with a head that pivots around an axis that is perpendicular to the blade. PCT Pub. No. WO 2006/017066A2/U.S. Pat. No. 7,497,860, PCT Pub. No. WO 2007/030793A2/U.S. Pat. No. 7,704,254, and PCT Pub. No. WO2016/182981A2/U.S. Pat. No. 10,687,823, the contents of which are each incorporated herein by reference, each disclose a sagittal saw blade cartridge. A sagittal saw blade cartridge includes a static guide bar and a blade head. The guide bar is an elongated member that is releasably attached to the handpiece, the saw that actuates the cartridge. The blade head is pivotally mounted to the guide bar and has teeth that extend forward from the guide bar. One or more drive links extend from the blade head towards the proximal end of the guide bar. The drive links are reciprocated back and forth by a drive assembly internal to the saw. The reciprocation of the drive links causes the blade head to pivot back and forth. The pivoting of the blade head is what enables the teeth to cut the tissue against which the blade head is pressed. Sometimes, this type of cartridge is referred to as an oscillating tip saw blade cartridge.

[0003] An advantage of the sagittal blade cartridge is that the only portion of the cartridge that pivots is the distally located blade head. By way of comparison, a conventional sagittal saw blade pivots from its point of attachment to the saw to which the blade is attached. A cartridge, when actuated, vibrates less in the hands of the surgeon holding the handpiece and/or a robotic arm. Also, it is common practice to use a cutting guide to properly position a sagittal saw blade relative to the tissue the blade is intended to cut. When a conventional blade is actuated, the oscillating movement of the blade imposes significant wear on the surfaces of the cutting guide defining the slot in which the blade is seated. The guide bar of a surgical sagittal blade cartridge only minimally moves in this slot. Thus, by using a cartridge, instead of a conventional blade, less of the material forming the cutting guide is rubbed off the guide.

[0004] When either a conventional sagittal blade or a sagittal blade cartridge advances through bone, the blade head is exposed to resistance and/or curved, angled, or shape surfaces. This resistance can be appreciable when the cut has a depth of 5 cm or more. Often the bone located adjacent the underside of the cartridge is more resistive to cutting than the bone located immediately above the cartridge. Similarly, the curve, angle, and/or shape of the bones surface can cause the sagittal blade cartridge to deflect in a certain direction. The cartridge, like most mechanical devices, will when advanced forward, advance along the path of least resistance. Since the bone above the cartridge can be less resistance to cutting than the bone below the cartridge, a cartridge when advanced, can flex upwardly out of the plane of the desired cut. This upwardly flexing or drifting of the blade is known as skiving. There can also be situations when owing to the density of the bone and to the shape of the bone as the point that the device enters, the cartridge, when advanced, flexes below the plane of the cut. This type of flexure or drifting of the blade is known as diving.

[0005] Regardless of the direction the blade flexes or drifts, it is undesirable. This is because a sagittal saw blade cartridge is typically used to remove bone so an artificial implant can be fitted in the space previously occupied by the removed bone. An implant is formed with surfaces designed to precisely seat against the complementary surfaces of the bone against which the implant is mounted. If the cut does not leave the bone with surfaces that have the desired shape, the results of the implant fitting procedure may be less than optimal.

[0006] In theory, one could increase the rigidity of a surgical sagittal blade cartridge by increasing the thickness of the guide bar. It should be appreciated that the slots of cutting guide through which the cartridge is inserted tend to be relatively narrow. Often the height of this slot is around 1.5 mm or less. This height limit imposes a limit of the thickness of the cartridge guide bar that can be inserted in this slot. Furthermore, if thickness of the guide bar is increased, by extension it is necessary to increase the thickness of the cut that will be formed by the cartridge. Increasing cut thickness can also lead to the cartridge leaving a cut surface that does not have the desired degree of planar smoothness. Increasing guide bar thickness is therefore typically not a viable solution for reducing the incidence of cartridge flexure. In robotic application cut guide thickness is not a limiting factor, but increasing the guide bar thickness thus requires an increase in the blade tip thickness, which also increase the amount of bone to be removed (more work to be done) which can be undesirable.

SUMMARY OF THE INVENTION

[0007] This invention is related to a new and useful surgical sagittal blade cartridge. The surgical sagittal blade cartridge includes a guide bar designed to resist deformation when subjected to uneven resistance.

[0008] The surgical sagittal blade cartridge of this invention includes a guide bar that is formed with three plates. There is a top plate and a bottom plate opposite the top plate. Between the top and bottom plate there is an inner plate. The inner plate is formed to have a curved head. The curved head functions as the boss around which the cartridge blade head pivots.

[0009] This Summary introduces a selection of concepts in a simplified form that are further described below in the Detailed Description below. This Summary is not intended to limit the scope of the claimed subject matter nor identify key features or essential features of the claimed subject matter.

[0010] In one aspect, a saw blade cartridge (**10**) for use with a surgical manipulator (**1**) is disclosed. The saw blade cartridge also includes a guide bar including a distal portion and a proximal portion, the guide bar may include: a first plate; a second plate; and an inner plate disposed partially between the first and the second plates, the inner plate defining a pivot surface. The cartridge also includes a blade at least partially disposed between the first and the second plates, the blade may include: The cartridge also includes a blade body including a proximal end and a distal end. The cartridge also includes a blade head disposed on the distal end of the blade body, the blade head formed with teeth, where the proximal end of the blade body abuts the pivot surface. The cartridge also includes where the proximal portion defines a proximal end. The cartridge also includes where one of the first and second plates define a first outer edge. The cartridge also includes where the inner plate defines a second outer edge, the second outer edge extends beyond the first outer edge defined by one of the first and second plates and where the second outer edge tapers outwardly from the proximal end to the distal end of the proximal portion such that the second outer edge assists in aligning the saw blade cartridge with the surgical device.

[0011] In a second aspect a surgical blade cartridge for use with a surgical manipulator is disclosed. The surgical blade cartridge also includes a first plate. The cartridge also includes a second plate. The cartridge also includes an inner plate, the inner plate defining a pivot surface. The cartridge also includes a blade at least partially disposed between the first and the second plates. The cartridge also includes a blade body including a proximal end and a distal end. The cartridge also includes a blade head disposed on the distal end of the blade body, the blade head formed with teeth, where the proximal end of the blade body abuts the pivot surface. The cartridge also includes the inner plate including a reference feature configured to facilitate determination of at least one of a position and/or orientation of the surgical blade to a navigation system. The cartridge also includes where the first plate, the second plate, and the inner plate are shaped to allow access to the reference feature on the inner plate.

[0012] In a third aspect, a method of registering a surgical saw blade with a navigation system is disclosed. The method also includes positioning an instrument including a first tracking device such that distal end of the instrument abuts the reference feature on the inner plate by inserting a distal end of the instrument through the opening in one of the first plate or the second plate. The method also includes determining an actual position and/or orientation of the reference feature based on the step of positioning. The method also includes determining an expected position and/or orientation of the reference feature based on predetermined geometrical data corresponding to the relationship between a mounting feature of a surgical manipulator and the reference feature of the surgical saw blade. The method also includes comparing the expected position and/or orientation of the reference feature with the actual position and orientation of the reference feature.

[0013] In a fourth aspect, a surgical sagittal saw blade cartridge is disclosed. The cartridge also includes a blade at least partially disposed between the first plate and the second plate at a distal end of the guide bar, the blade may include: a blade body including a proximal end and a distal end; and a blade head disposed on the distal end of the blade body and extending from the distal end of the guide bar, the blade head formed with teeth, the teeth may include: a plurality of base teeth; and a first pilot tooth. The cartridge also includes at least one drive link moveably disposed within a space defined between the first plate and the second plate of the guide bar, the at least one drive link having opposed proximal and distal ends, the proximal end adapted to be attached to a drive element of a surgical device and the distal end being connected to the proximal end of the blade body so that reciprocation of the at least one drive link results in pivoting of the blade about a pivot surface. The cartridge also includes where the base teeth are configured to move along a first arc when the blade pivots, and the first pilot tooth is configured to move along a second arc. The cartridge also includes where the second arc is distal to the first arc.

[0014] In a fifth aspect, a surgical blade of a sagittal saw blade cartridge including a proximal end and a distal end and defining a pivot surface between the proximal end and the distal end is disclosed. The surgical blade also includes a blade body including a proximal end a distal end, the blade body configured to pivot about the pivot surface. The blade also includes a blade head disposed on the distal end of the blade body and extends from the distal end of the sagittal saw blade cartridge, the blade head formed with teeth, the teeth may include: a plurality of base teeth, a first pilot tooth, and a second pilot tooth. The blade also includes a first plane (plane-**1**) extending between the proximal end the distal end of the blade body, the first plane oriented generally orthogonal to a top surface of the blade body and bisects the blade body; a second plane (plane-**2**) perpendicular to the first plane, the second plane bisecting the blade body to define a top portion and a bottom portion; where the intersection of the first plane (plane-**1**) and the second plane (plane-**2**) defines a central axis of the blade body; and a mirroring plane (plane-**3**) configured to intersect the central axis and oriented at a mirror angle (θ) measured from the second plane (plane-**2**). The blade also includes where the blade head is configured such that the second pilot tooth is a mirrored representation of the first pilot tooth mirrored about the mirroring plane (plane-**3**).

[0015] In a sixth aspect, a surgical blade including a proximal end and a distal end and defining a pivot surface between the proximal end and the distal end is disclosed. The surgical blade also includes a blade body including a proximal end a distal end, the blade body configured to pivot about the pivot surface. The blade also includes a blade head disposed on the distal end of the blade body, the blade head formed with teeth, the teeth may include: a plurality of base teeth, and a first pilot tooth. The blade also includes a first plane extending between the proximal end the distal end of the blade body, the first plane oriented generally orthogonal to a top surface of the blade body and bisects the blade body; a second plane perpendicular to the first plane, the second plane bisecting the blade body to define a first portion and a second portion; where the first pilot tooth defines a first apex relative to the pivot surface. The blade also includes where the base teeth define a second apex relative to the pivot surface. The blade also includes the first apex of the first pilot tooth is distal to the second apex of the base teeth.

[0016] In a seventh aspect, a method of cutting biological tissue using a surgical blade including a blade head may include a pilot tooth that extends distally beyond a plurality of adjacent base teeth. The method of cutting biological tissue also includes automatically controlling a surgical manipulator to position the surgical blade on a target plane relative to the biological tissue such that the first pilot tooth will contact the biological tissue prior to any of the plurality of base teeth contacting the biological tissue; actuating the surgical blade using a surgical manipulator while the blade is on the target plane.

[0017] In an eighth aspect, a surgical sagittal saw blade cartridge for use with a surgical device is disclosed. The surgical sagittal saw blade cartridge also includes a guide bar including a distal portion and a proximal portion, the guide bar may include: a first plate; a second plate; and an inner plate disposed between the first and the second plates, the inner plate defining a pivot surface, a first opening. The cartridge also includes a blade at least partially disposed between the first plate and the second plate at a distal end of the guide bar, the blade having a blade body defining a second pivot surface face to pivot about the pivot surface of the inner plate. The cartridge also includes where one of the first and second plates define a first outer edge of the proximal portion of the guide bar. The cartridge also includes where the inner plate defines a second outer edge of the proximal portion of the guide bar and the guide bar is formed such that the second outer edge extends beyond the first outer edge defined by one of the first and second plates. The cartridge also includes where the combination of the first opening of the inner plate and the second outer edge of the proximal portion of the guide bar defined by the inner plate are configured to axially align the saw blade cartridge relative to the surgical device.

[0018] In a ninth aspect, a surgical sagittal saw blade cartridge for use with a surgical device is disclosed. The surgical sagittal saw blade cartridge also includes a guide bar including a distal portion and a proximal portion, the guide bar may include: a first plate; a second plate; and an inner plate disposed between the first and the second plates, the inner plate defining a pivot surface, a first opening, and a second opening. The cartridge also includes a longitudinal axis extending between the distal portion and the proximal portion of the guide bar, the longitudinal axis configured to bisect the guide bar. The cartridge also includes a blade at least partially disposed between the first plate and the second plate at a distal end of the guide bar, the blade having a blade body defining a pivot surface to pivot about the pivot surface of the inner plate. The cartridge also includes where one of the first and second plates define a first outer edge of the proximal portion of the guide bar. The cartridge also includes where the inner plate defines a second outer edge of the proximal portion of the guide bar and the guide bar is formed such that the second outer edge extends beyond the first outer edge defined by one of the first and second plates. The cartridge also includes where each of the first opening, and the second opening are positioned along the longitudinal axis. The cartridge also includes where the combination of the first opening and the second opening of the inner plate are configured to axially align the saw blade cartridge relative to the surgical device.

[0019] In a tenth aspect, a surgical blade cartridge for use with a surgical device is disclosed. The cartridge also includes a second plate. The cartridge also includes an inner plate, the inner plate defining a pivot surface, a first opening, and a reference feature, the reference feature configured to facilitate determination of at least one of a position and/or orientation of the surgical blade to a navigation system. The cartridge also includes where at least of the first plate or the second plate defines an opening configured to provide access to the reference feature on the inner plate. The cartridge also includes where the first opening in the inner plate is configured to axially align a center of a radius of the pivot surface relative to the surgical device.

[0020] In an eleventh aspect, a method of validating installation of a surgical saw blade relative to a surgical manipulator to which the surgical saw blade is coupled through a mounting feature is disclosed. The method of validating installation also includes tracking, with the navigation system, the first tracking device as the distal tip of the instrument is placed to abut the reference feature. The installation also includes determining, with the navigation system, an actual position and orientation of the reference feature based on the step of tracking. The installation also includes comparing, with the navigation system, the actual position and orientation of the reference feature to an expected position and orientation of the reference feature, the expected position and orientation based on predetermined geometrical data corresponding to a relationship between the mounting feature of the surgical manipulator and the reference feature of the surgical saw blade.

[0021] In a twelfth aspect, a method of attaching a surgical saw blade to a surgical manipulator is disclosed. The method also includes positioning the surgical saw blade such that an alignment post of the surgical manipulator is located the first opening and a coupling post of the surgical manipulator is located in the second opening of the inner plate. The method also includes axially aligning the inner plate relative to the surgical manipulator by sliding the inner plate proximally toward the surgical manipulator causing each of the alignment post and the coupling post to move from a proximal position within each of the first opening and the second opening respectively to a distal position within each of the first opening and the second opening.

[0022] In a thirteenth aspect, a method assembling a surgical sagittal saw is disclosed. The method also includes attaching a surgical saw blade to a surgical manipulator by positioning the saw surgical saw blade such that an alignment post of the surgical manipulator is located in a first opening of the surgical saw blade and the coupling post of the surgical manipulator is located in a second opening of the surgical saw blade. The method also includes sliding the surgical saw blade proximally toward the surgical manipulator causing each of the alignment post and the coupling post to move from a proximal position within each of the first opening and the second opening respectively to a distal position within each of the first opening and the second opening. The method also includes where the alignment post is sized to simultaneously engage opposing lateral surfaces of the first opening when positioned within the distal position of the first opening to axially align the surgical saw blade relative to the surgical manipulator as the surgical saw blade is slid proximally toward the surgical manipulator.

[0023] In a fourteenth aspect, a surgical sagittal saw blade cartridge for use with a surgical device is disclosed. The surgical sagittal saw blade cartridge also includes a guide bar including a distal portion and a proximal portion, the guide bar may include: a first plate; a second plate; and an inner plate disposed between the first and the second plates, the inner plate including an inner tine defining a pivot surface. The cartridge also includes a longitudinal axis extending between the distal portion and the proximal portion of the guide bar, the longitudinal axis configured to bisect the guide bar. The cartridge also includes a blade at least partially disposed between the first plate and the second plate at a distal end of the guide bar, the blade having a blade body defining a pivot face to pivot about the pivot surface of the inner tine. The cartridge also includes where one of the first and second plates define a first outer edge of the proximal portion of the guide bar. The cartridge also includes where the inner plate defines a second outer edge of the proximal portion of the guide bar and the guide bar is formed such that the second outer edge extends laterally beyond the first outer edge defined by one of the first and second plates relative to the longitudinal axis. The cartridge also includes where at least one of the first and the second plates are coupled to the inner plate by a weld formed between the inner tine the first and/or the second plates.

[0024] In a fifteenth aspect, a saw blade cartridge for use with a surgical device is disclosed. The saw blade cartridge also includes a guide bar including a distal portion and a proximal portion, the guide bar may include: a first plate, a second plate, the first plate coupled to the second plate and the first plate including a pivot surface. The cartridge also includes a blade at least partially disposed between the first and the second plates, the blade may include: a blade body including a proximal end and a distal end; and a blade head disposed on the distal end of the blade body and configured to extend from the distal portion of the guide bar, the blade head formed with teeth, where the proximal end of the blade body abuts the pivot surface. The cartridge also includes where the proximal portion defines a proximal end. The cartridge also includes where the first plate defines a first outer edge. The cartridge also includes where the second plate defines a second outer edge. The cartridge also includes the guide bar is configured such that the first outer edge extends beyond the second outer edge. The cartridge also includes where the first outer edge tapers outwardly from the proximal end to the distal end of the proximal portion of the guide bar such that the first outer edge assists in aligning the saw blade cartridge with the surgical device.

[0025] In a sixteenth aspect, a saw blade cartridge for use with a surgical device is disclosed. The saw blade cartridge also includes a guide bar including a distal portion and a proximal portion, the guide bar may include: a first plate; a second plate; and an inner plate disposed partially between the first and the second plates, the inner plate defining a pivot surface. The cartridge also includes a blade at least partially disposed between the first and the second plates, the blade may include: The cartridge also includes a blade body including a proximal end and a distal end. The cartridge also includes a blade head disposed on the distal end of the blade body and configured to extend from the distal portion of the guide bar, the blade head formed with teeth, where the proximal end of the blade body abuts the pivot surface. The cartridge also includes a longitudinal axis extending between the distal portion and the proximal portion of the guide bar. The cartridge also includes where the proximal portion defines a proximal end. The cartridge also includes where one of the first and second plates define a first outer edge of the proximal portion of the guide bar. The cartridge also includes where the inner plate is configured to define a second outer edge of the proximal portion of the guide bar, the second outer edge extends beyond the first outer edge defined by one of the first and second plates.

[0026] In a seventeenth aspect, a saw blade cartridge for use with a surgical device is disclosed. The saw blade cartridge also includes a guide bar including a distal portion and a proximal portion, the guide bar may include: a first plate; a second plate; and an inner plate disposed partially between the first and the second plates, the inner plate configured to define a pivot surface. The cartridge also includes a blade at least partially disposed between the first and the second plates. The cartridge also includes a longitudinal axis extending between the distal portion and the proximal portion of the guide bar. The cartridge also includes where the proximal portion defines a proximal end. The cartridge also includes where one of the first and second plates define a first outer edge. The cartridge also includes where the inner plate defines a second outer edge, the second outer edge extends beyond the first outer edge defined by one of the first and second plates and where the second outer edge tapers outwardly from the proximal end to the distal end of the proximal portion such that the second outer edge assists in aligning the saw blade cartridge with the surgical device. The cartridge also includes where the inner plate further may include a reference feature configured to facilitate determination of at least one of a position and/or orientation of the surgical blade with a navigation system. The cartridge also includes where at least one of the first plate or the second plate is shaped to provide access to the reference feature of the inner plate.

[0027] In an eighteenth aspect, a surgical saw system is disclosed. The surgical saw system also includes a surgical device including a mount, the mount defining a mount surface, and the mount may include: a coupling post, and an alignment post. The system also includes a saw blade cartridge removably couplable to the mount of the surgical device, the saw blade cartridge may include: a guide bar including a distal portion and a proximal portion, the guide bar may include: an inner plate defining a pivot surface, a first opening, and a second opening. The system also includes a blade disposed at a distal end of the guide bar, the blade having a blade body defining a pivot face to pivot about the pivot surface of the inner plate. The system also includes where the first opening is sized to receive and slidably engage the alignment post of the mount to axially align the saw blade cartridge relative to the surgical device. The system also includes where the second opening is sized to receive the coupling post of the mount to removably couple the saw blade cartridge to the surgical device.

[0028] In a nineteenth aspect, a surgical saw system is disclosed. The surgical saw system also includes a surgical device including a mount, the mount defining a mount surface, and the mount may include: a first wall and a second wall extending above the mounting surface, the first and second wall positioned on opposing sides of a longitudinal axis of the mount; the first and second wall are tapered relative to the longitudinal axis of the mount to facilitate alignment of the saw blade cartridge along the longitudinal axis; at least one tab disposed on each of the first and second walls, each of the at least one tabs positioned above and extending over the mounting surface. The system also includes a saw blade cartridge removably couplable to the mount of the surgical device, the saw blade cartridge may include: a guide bar including a distal portion and a proximal portion, the guide bar may include: a first plate; a second plate; and an inner plate disposed partially between the first and the second plates. The system also includes where one of the first and second plates define a first outer edge. The system also includes where the inner plate is configured to define a second outer edge, the second outer edge extends beyond the first outer edge defined by one of the first and second plates; where each of the first and second walls engage the second outer edge of the inner plate and an under surface of each of the at least one tabs engage a top surface of the inner plate.

[0029] In a twentieth aspect, a saw blade cartridge for use with a surgical device. The saw blade cartridge also includes a guide bar including a distal portion and a proximal portion, the guide bar may include: a first plate, a second plate. The cartridge also includes a blade at least partially disposed between the first and the second plates. The cartridge also includes a blade body including a proximal end and a distal end. The cartridge also includes a blade head disposed on the distal end of the blade body and configured to extend from the distal portion of the guide bar, the blade head formed with teeth; where the blade head defines one or more apertures for allowing debris to exit.

[0030] In a twenty-first aspect, a method of manufacturing a saw blade cartridge for use with a surgical manipulator is disclosed. The method also includes forming one or more apertures in a blank piece of material to be a blade head of the saw blade cartridge; and machining the blank to form one or more teeth at a location on the blank based on the location and/or position of the one or more apertures.

[0031] In a twenty-second aspect, a surgical sagittal saw blade cartridge for use with a surgical device is disclosed. The surgical sagittal saw blade cartridge also includes a guide bar including a distal portion and a proximal portion, the guide bar may include: a first plate, a second plate, and one of the first and second plate defining a pivot member. The cartridge also includes a longitudinal axis extending between the distal portion and the proximal portion of the guide bar, the longitudinal axis configured to bisect the guide bar. The cartridge also includes a blade at least partially disposed between the first plate and the second plate at a distal end of the guide bar, the blade having a blade body defining a pivot surface to pivot about the pivot surface. The cartridge also includes where the guide bar has a first outer edge of the proximal portion of the guide bar and a second outer edge of the proximal portion of the guide bar and the guide bar is formed such that the second outer edge extends beyond the first outer edge. The cartridge also includes the one of the plates that defines the pivot member include a first opening positioned along the longitudinal axis. The cartridge also includes where the combination of the first opening and the second outer edge are configured to axially align the saw blade cartridge relative to the surgical device.

[0032] In a twenty-third aspect, a surgical sagittal saw blade cartridge for use with a surgical device is disclosed. The surgical sagittal saw blade cartridge also includes a guide bar including a distal portion and a proximal portion, the guide bar may include: a first plate, a second plate, and one of the first and second plate defining a pivot member. The cartridge also includes a longitudinal axis extending between the distal portion and the proximal portion of the guide bar, the longitudinal axis configured to bisect the guide bar. The cartridge also includes a blade at least partially disposed between the first plate and the second plate at a distal end of the guide bar, the blade having a blade body defining a pivot surface to pivot about the pivot surface. The cartridge also includes where the guide bar has a first outer edge of the proximal portion of the guide bar and a second outer edge of the proximal portion of the guide bar and the guide bar is formed such that the second outer edge extends beyond the first outer edge. The cartridge also includes where the second outer edge is configured to axially align the saw blade cartridge relative to the surgical device.

[0033] In a twenty-fourth aspect, a saw blade cartridge for use with a surgical device and registration with a surgical navigation system using a registration tool is disclosed. The saw blade cartridge also includes a guide bar including a distal portion and a proximal portion, the guide bar may include: a first plate; a second plate; and an inner plate disposed partially between the first and the second plates; one of the first and second plates defining an opening to provide access to a surface of the inner plate, the opening shaped to define a first region and a second region; the first region having a first size and being configured to receive insertion of a registration tool; and the second region having a second size that is less than the first size and configured to position the registration tool at a reference feature on the guide bar and/or the inner plate. The cartridge also includes a blade at least partially disposed between the first and the second plates.

[0034] In a twenty-fifth aspect, a saw blade cartridge for use with a surgical device is disclosed. The saw blade cartridge also includes a guide bar including a distal portion and a proximal portion, the guide bar may include: a first plate; a second plate; and an inner plate disposed partially between the first and the second plates, the inner plate configured to define a pivot surface. The cartridge also includes a blade at least partially disposed between the first and the second plates. The cartridge also includes a blade body including a proximal end and a distal end. The cartridge also includes a blade head disposed on the distal end of the blade body and configured to extend from the distal portion of the guide bar, the blade head formed with teeth, where the proximal end of the blade body abuts the pivot surface. The cartridge also includes where a surface of the inner plate defines a contact point to removably engage the surgical device. The cartridge also includes where one of the first and second plates is shaped to provide access to the contact point on the inner plate, such that the contact point on the inner plate assists in aligning the saw blade cartridge with the surgical device.

[0035] In a twenty-sixth aspect, a surgical sagittal saw blade cartridge for use with a surgical device is disclosed. The surgical sagittal saw blade cartridge also includes a guide bar including a distal portion and a proximal portion, the guide bar may include: a first plate; a second plate; and an inner plate disposed between the first and the second plates, the inner plate defining a pivot surface, a first opening. The cartridge also includes a blade at least partially disposed between the first plate and the second plate at a distal end of the guide bar, the blade having a blade body defining a second pivot surface face to pivot about the pivot surface of the inner plate. The cartridge also includes where one of the first and second plates define a first outer edge of the proximal portion of the guide bar. The cartridge also includes where the inner plate defines a second outer edge of the proximal portion of the guide bar and the guide bar is formed such that the second outer edge extends beyond the first outer edge defined by one of the first and second plates. The cartridge also includes where at least one of the first opening of the inner plate and the second outer edge of the proximal portion of the guide bar defined by the inner plate are configured to orient the saw blade cartridge relative to the surgical device.

[0036] In some of the implementations described above, the guide bar may further defines a longitudinal axis extending between the distal portion and the proximal portion of the guide bar; and wherein the second outer edge defines a first taper angle relative to the longitudinal axis

between a first point and a second point on the second outer edge; and wherein the second outer edge defines a second taper angle relative to the longitudinal axis between the second point and a third point on the second outer edge, the first point being closer to a proximal end of the cartridge than the third point and the second point being disposed between the first point and the third point.
[0037] In some of the implementations described above, the second outer edge defines a third taper angle relative to the longitudinal axis between the third point and a fourth point on the second outer edge.
[0038] In some of the implementations described above, the pivot surface may comprise an arcuate surface, and the proximal end of the blade body defining a recess that abuts the pivot surface, such that the blade body is configured to pivot about the pivot surface defined by the inner plate when the blade body is actuated.
[0039] In some of the implementations described above, the inner plate may further define a slot configured to axially align the guide bar relative to the surgical device, and the axial slot may be oriented such that the longitudinal axis of the slot is parallel to the longitudinal axis of the guide bar.
[0040] In some of the implementations described above, at least one of the plates may define a reference feature configured to facilitate determination of a position and/or orientation of the surgical blade with a navigation system, and the inner plate further comprises a reference feature configured to facilitate determination of a position and/or orientation of the surgical blade with a navigation system.
[0041] In some of the implementations described above, the reference feature is selected from the group comprising an optical marking, a divot, or an aperture, the reference feature comprises a plurality of laser markings, and wherein at least one of the plurality of laser markings is positioned on each side of the longitudinal axis.
[0042] In some of the implementations described above, at least one of the first plate or the second plate is shaped to provide access to the reference feature of the inner plate, and the reference features is disposed on a surface of the inner plate that is exposed by the second outer edge of the inner plate extending beyond the first outer edge defined by one of the first and second plates.
[0043] In some of the implementations described above, the reference feature is configured to identify the saw blade cartridge to the navigation system, including one or more characteristics of the saw blade cartridge, and the characteristic including at least one of the following: a type of the blade, a blade thickness of the blade head, a tooth configuration of the blade head, a length of the guide bar, and/or a width of the guide bar.
[0044] In some of the implementations described above, the surgical manipulator and/or the surgical device is one of a hand-held tool, a robotic arm, or a hand-held robot.
[0045] In some of the implementations described above, the guide bar is further formed such that the second outer edge extends laterally beyond the first outer edge defined by one of the first and second plates relative to a longitudinal axis extending between the distal portion and the proximal portion of the guide bar.
[0046] In some of the implementations described above, the inner tine further defines a first opening, the first opening configured to align the pivot surface relative to the surgical device.
[0047] In some of the implementations described above, the guide bar may be configured such at least one of the first opening of the inner plate and the second outer edge of the proximal portion of the guide bar defined by the inner plate are configured to orient the saw blade cartridge relative to the surgical device.
[0048] In some of the implementations described above, wherein a surface of the inner plate defines a contact point to removably engage the surgical device; wherein one of the first and second plates is shaped to provide access to the contact point on the inner plate, such that the contact point on the inner plate assists in aligning the saw blade cartridge with the surgical device.
[0049] In some of the implementations described above, the guide bar may be configured one of

the first and second plates defining an opening to provide access to a surface of the inner plate, the opening shaped to define a first region and a second region; the first region having a first size and being configured to receive insertion of a registration tool; and the second region having a second size that is less than the first size and configured to position the registration tool at a reference feature on the guide bar and/or the inner plate.
[0050] In some of the implementations described above, the blade head may define one or more apertures for allowing debris to exit and or to be used as a reference point when manufacturing the blade.
[0051] Any of the above aspects can be combined in full or in part. Any features of the above aspects can be combined in full or in part. Any of the above implementations for any aspect can be combined with any other aspect. Any of the above implementations can be combined with any other implementation whether for the same aspect or different aspect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0052] The invention is pointed out with particularity in the claims. The above and further features and benefits of this invention are understood from the following Detailed Description taken in conjunction with the accompanying drawings in which:
[0053] FIG. **1** is a top view of a first configuration of a saw blade cartridge including a distal portion with a single taper.
[0054] FIG. **2** is a top view of the distal portion of the saw blade cartridge of FIG. **1** including a first configuration of a saw blade, the first configuration saw blade including a blade head with a plurality of base teeth and two pilot teeth.
[0055] FIG. **3** is a side view of the saw blade cartridge of FIG. **1**.
[0056] FIG. **4** is a side view of the distal portion of the saw blade cartridge of FIG. **1** illustrating a side profile of blade head and the pilot teeth.
[0057] FIG. **5** is a top view of the blade head of the first configuration of the saw blade of the saw blade cartridge of FIG. **1**, the blade head including two pilot teeth proximate the central axis off the saw blade and configured to extend distally of the base teeth of the blade head.
[0058] FIG. **6** is a front view of the blade head of the saw blade of the saw blade cartridge of FIG. **1**.
[0059] FIG. **7** is a front view of the blade head of the first configuration of the saw blade of the saw blade cartridge of FIG. **1** illustrating a front profile of the base teeth and pilot teeth.
[0060] FIG. **8** is a top sectional view illustrating the interior features of the
[0061] saw blade cartridge of FIG. **1**.
[0062] FIG. **9** is an exploded view of the saw blade cartridge of FIG. **1**.
[0063] FIG. **10** is a front perspective view of the distal portion of the saw blade cartridge of FIG. **1**.
[0064] FIG. **11** is a side perspective view of the distal portion of the saw blade cartridge of FIG. **1**.
[0065] FIG. **12** is a top view of a distal portion of the saw blade cartridge of FIG. **1** including a second configuration of a saw blade, the second configuration of the saw blade including a blade head with a plurality of base teeth and a single pilot tooth.
[0066] FIG. **13** is a top view of the blade head of the second configuration of the saw blade of FIG. **12**, the blade head including the single pilot tooth proximate the central axis off the saw blade and configured to extend distally of the base teeth of the blade head.
[0067] FIG. **14** is a front view of the blade head of the second configuration of the saw blade of FIG. **12** illustrating a front profile of the base teeth and pilot tooth.
[0068] FIG. **15** is a side view of the distal portion of the second configuration of the saw blade of FIG. **12** illustrating a side profile of the blade head and the pilot tooth.

[0069] FIG. **16** is a top view of a second configuration of a saw blade cartridge including a distal portion with multiple tapers and a third configuration of a saw blade including a blade head with a plurality of base teeth and a single pilot tooth.
[0070] FIG. **17** is a side view of the saw blade cartridge of FIG. **16**.
[0071] FIG. **18** is a side view of the distal portion of the saw blade cartridge of FIG. **16** illustrating a side profile of blade head including the single pilot tooth.
[0072] FIG. **19** is a top view of the distal portion of the saw blade cartridge of FIG. **16**.
[0073] FIG. **20** is a top view of the blade head of the second configuration of the saw blade of FIG. **16**, the blade head including the single pilot tooth proximate the central axis off the saw blade and configured to extend distally of the base teeth of the blade head.
[0074] FIG. **21** is a front view of the blade head of the third configuration of the saw blade of the saw blade cartridge of FIG. **16** illustrating a front profile of the base teeth and pilot tooth.
[0075] FIG. **22** is a side perspective view of the distal portion of the saw blade cartridge of FIG. **16**.
[0076] FIG. **23** is a top section al view illustrating the interior features of the saw blade cartridge of FIG. **16**.
[0077] FIG. **24** is an exploded view of the saw blade cartridge of FIG. **16**.
[0078] FIG. **25** is a perspective view of an exemplary configuration of a mount for a saw blade cartridge.
[0079] FIG. **26** is a top view of the mount for a saw blade cartridge of FIG. **24**.
[0080] FIG. **27** is a front perspective view of the mount for a saw blade cartridge of FIG. **24**.
[0081] FIG. **28** is a perspective view of the saw blade cartridge of FIG. **1** mounted to the mount of FIG. **24**.
[0082] FIG. **29** is a top view of the saw blade cartridge of FIG. **1** mounted to the mount of FIG. **24**.
[0083] FIG. **30** is a rear perceptive view of the saw blade cartridge of FIG. **1** mounted to the mount of FIG. **24** illustrating the interaction between the mounting features of the proximal portion of the saw blade cartridge with the mount.
[0084] FIG. **31** is a perspective view of a sectional view of the saw blade cartridge of FIG. **1** mounted to the mount of FIG. **24** illustrating how the interior features of the saw blade cartridge interact with the mount.
[0085] FIG. **32** is a perceptive view of the saw blade cartridge of FIG. **1** mounted to the mount of FIG. **24**, the mount connected to a surgical robot used to operate the saw blade cartridge.
[0086] FIG. **33** is a top view of the saw blade cartridge of FIG. **1** including a fourth configuration of a saw blade and an inner plate including a plurality of reference features.
[0087] FIG. **34** is an exploded view of the saw blade cartridge of FIG. **34**.
[0088] FIG. **35**A is a top view of the distal portion of the saw blade cartridge of FIG. **33** including the fourth configuration of a saw blade, the fourth configuration saw blade including at least one aperture in the saw blade.
[0089] FIG. **35**B is a top view of the distal portion of the saw blade cartridge of FIG. **33** including a fifth configuration of a saw blade, the fifth configuration saw blade including a plurality of apertures in the saw blade.
[0090] FIG. **36** is a top sectional view illustrating the interior features of the saw blade cartridge of FIG. **33**, including the blade with apertures and the inner plate including reference features.
[0091] FIG. **37** is a top view of the saw blade cartridge of FIG. **1** including a second configuration of a guide bar including an opening in at least one of a first plate or a second plate, the opening sized and/or shaped to guide a registration tool to a reference feature on the guide bar.
[0092] FIG. **38** is an exploded view of the saw blade cartridge of FIG. **37**.
[0093] FIG. **39** is a top view of the saw blade cartridge of FIG. **1** including a third configuration of a guide bar including an opening in at least one of a first plate or a second plate, the opening sized and/or shaped to guide a registration tool to a reference feature on the guide bar
[0094] FIG. **40** is an exploded view of the saw blade cartridge of FIG. **39**.

DETAILED DESCRIPTION

[0095] FIGS. **1** through **11** depict a first configuration of a blade cartridge **10**. The sagittal blade cartridge may also be referred to as a sagittal blade cartridge, surgical blade cartridge, or cartridge. The blade cartridge **10** may comprises a guide bar **11**. The guide bar **11** may comprises a plurality of plates **12**, **14**, **16**, including a first plate **12**, a second plate **14**, and a third plate **16**. The guide bar **11** may be configured such that the third plate **16** may be disposed between the first plate **12** and the second **14**. As such, the first plate **12** may also be referred to as the bottom plate, the second plate **14** may be referred to as a top plate, and the third plate **16** may be referred to as an inner plate, center plate, inner member, center layer, and/or center member. While the guide bar **11** is illustrated as only including three plates layered in a laminate type arrangement, it is contemplated that the guide bar **11** may comprise any number of plates and/or layers.

[0096] Collectively, the plates are formed so that the guide bar **11** has a proximal portion **18** that forms the proximal end **18**A, **18**B, **18**C of the plates **12**, **14**, **16**, The proximal portion **18** of the guide bar **11** may be configured to initially taper outwardly as you move distally along the guide bar **11**, and then eventually taper inwardly as you move further distally along the guide bar **11**. The outward taper of the proximal portion **18** of the guide bar **11** may configured to as a mounting feature to assist in mounting and/or securing the saw blade cartridge **10** to a surgical manipulator **301**, such as a surgical robot or medical handpiece. For example, as illustrated in FIG. **1**, opposed outer edges of the proximal portion **18**B of the third plate **16** may extend beyond the opposed outer edges **36**, **86** of the proximal portion **18**A, **18**C of the first and/or second plates **12**, **14**. It is a contemplated that the entire length of the opposed outer edges of the of the proximal portion **18**B of the third plate **16** may extend beyond on the opposed outer edges **36**, **86** defined by the proximal portion **18**A, **18**C of the first and/or second plates **12**, **14**. Alternatively, it is also contemplated that the guide bar **11** may be configured such that only a portion of the opposed outer edges of the of the proximal portion **18**B of the third plate **16** may extend beyond on the opposed outer edges **36**, **86** defined by the proximal portion **18**A, **18**C of the first and/or second plates **12**, **14**. For example, referring back to FIG. **1**, only a portion of the opposed outer edges of the of the proximal portion **18**B of the third plate **16** extend beyond on the opposed outer edges **36**, **86** defined by the proximal portion **18**A, **18**C of the first and/or second plates **12**, **14**. While not illustrated in the figures, it is further contemplated that one or both of the proximal portion(s) **18**A, **18**C of the first and/or second plates **12**, **14** may be shaped to define a recess or include a cutout defined in the outer edges **36**, **86** of the proximal portion **18**A, **18**C of the first and/or second plates **12**, **14** that exposes a portion of the proximal portion **18**B of the third plate **16** to be engaged by and/or contact the mount **303** of the surgical manipulator **301**. It is also contemplated that only one of the outer edges **36**, **86** of the proximal portion **18**A, **18**C of the first and/or second plates **12**, **14** may be shaped to expose a portion of the proximal portion **18**B of the third plate **16** to be engaged by and/or contact the mount **303** of the surgical manipulator **301**. In this configuration, the other of the outer edges **36**, **86** of the proximal portion **18**A, **18**C of the first and/or second plates **12**, **14** may be generally sized and/or shaped to mimic the size and shape of the outer edge of the proximal portion **18**B of the third plate **16**. The opposed edges of the proximal portion **18**B of the third plate **16** extending beyond the opposed outer edges **36**, **86** of the proximal portion **18**A, **18**C of the first and/or second plates **12**, **14** may allow for the outer edge of the proximal end **18**B of the third plate to be the portion of the guide bar that contacts and/or interacts with the mount **303** of the surgical manipulator **301** for the purpose of mounting and/or aligning of the saw blade cartridge **10** relative to the surgical manipulator **301**. This will be described in greater detail below.

[0097] Referring to FIG. **8**, the opposed outer edges of the proximal portion **18**B of the third plate **16** may comprise a number of points **39**, **41**, **43**, **45** spaced along the outer edge of the proximal portion **18**. The points **39**, **41**, **43**, **45** along the edge of the proximal portion **18** of the third plate **16** may define a plurality of segments **40**, **42**, **44** of the edge of the proximal portion **18** of the third plate **16**, each of which may comprise varying degrees of taper relative to a longitudinal axis L that extends from the proximal to the distal end of the guide bar **11**. For example, the edge of the proximal portion **18**B of the third plate **16** may comprise a first segment **40** defined between a first point **39** and a second point **41**. The first segment **40** may be generally parallel to the longitudinal axis L of the guide bar **11**. Alternatively, the first segment **40** may comprise a first outward directed taper such that the second point **41** is further from the longitudinal axis L of the guide bar **11** than the first point **39**.

[0098] The outer edge of the proximal portion **18**B of the third plate **16** may comprise a second segment **42** defined between the second point **41** and a third point **43**. The second segment **42** may comprise a second outward directed taper such that the third point **43** is further from the longitudinal axis L of the guide bar **11** than the second point **41**. The outer edge of the proximal portion **18** of the third plate **16** may comprise a third segment **44** defined between the third point **43** and a fourth point **45**. The third segment **44** may comprise a third outward directed taper such that the fourth point **45** is further from the longitudinal axis L of the guide bar **11** than the third point **43**. The outer edge of the proximal portion **18**B of the third plate **16** may comprise a second segment **42** defined between the second point **41** and a third point **43**. The second segment **42** may comprise a second outward directed taper such that the third point **43** is further from the longitudinal axis L of the guide bar **11** than the second point **41**. The proximal portion **18**B of the third plate **16** may be configured such that each of the first, second, and third segments **40**, **42**, **44** comprise a different taper relative to the longitudinal axis L of the guide bar **11**. For example, the first segment **40** may be generally parallel to the longitudinal axis L, the second segment **42** may comprise the second taper, and the third segment **44** may comprise the third taper wherein the second taper extends outward at a larger angle relative to the longitudinal axis L than the third taper. The taper and/or orientation of each of the segment **40**, **42**, **44** of the opposed outer edges of the proximal portion **18**B of the third plate **16** may be correspond to and/or be matched to an angle and/or taper of a surface of a bracket **317** of a mount **303** for securing the saw blade cartridge **10** to the surgical manipulator **301**, which will be described in greater detail below.

[0099] The proximal portion **18**B of the third plate **16** may further comprise a fourth segment **47** of the opposed outer edges that is distal of the third segment **44**. The fourth segment **47** of the opposed outer edges of the proximal portion **18**B may be configured to be generally parallel to the longitudinal axis L of the guide bar. It is also contemplated that the fourth segment **47** of the opposed edges of the proximal portion **18**B may be tapered outwardly as you move distally along the guide bar **11**, or it may be tapered inwardly as you move distally along the guide bar **11**. The proximal portion **18**B of the third plate **16** may further comprise a fifth segment **49** of the opposed edges that is distal of the fourth segment **44**. The fifth segment **49** of the opposed edges of the proximal portion **18**B may be configured to be tapered inwardly as you move distally along the guide bar **11**.

[0100] Forward of the proximal portion **18** of the guide bar **11** is a middle portion and a distal portion **20**. The plates **12**, **14**, **16** are formed so that the opposed sides of the guide bar that form the sides of the middle portion are generally parallel. The opposed sides of the distal portion **20** of the guide bar **11** may be tapered in some configurations. As illustrated in the figures, opposed edges **32**, **99** of the distal portion **20**A, **20**B, **20**C of the guide bar **11** may be tapered inwardly toward the center of the plates **12**, **14**, **16**. It is also contemplated that the opposed edges of the distal portion **20**A, **20**B, **20**C of the guide bar **11** may be generally parallel to one another. The width across the middle portion and/or the distal portion **20** of the guide bar **11** may be less that the width of the proximal portion **18** at the widest point between the opposed edges of the proximal portion **18** of the guide bar **11**.

[0101] The proximal portion **18** of each of the plates **12**, **14**, **16** may further comprise two laterally spaced apart openings **22**, **88** (one identified). As illustrated in the figures, the first plate may comprise a pair of openings **22**, and the second plate **14** may comprise a pair of openings **88**. The third plate **16**, depending on the shape of the plate **16** may or may not define an opening. As illustrated in the figures, the third plate **16** is shaped to define a channel **50**A, **50**B that cooperates with the openings **22**, **88** defined in the first and second plates **12**, **14**. While the plates are illustrated including a pair of openings **22**, **88** defined in the first and second plates **12**, **14**, and a pair of channels **50**A, **50**B defined by the third plate **16**, it is contemplated that the guide bar **11** may be formed with a single opening in each of the plates **22**, **88**, as well as with more than two opening **22**, **88** defined in each of the plates **12**, **14**, **16**. The openings **22**, **88** may formed to have a semi-oval shape. Each opening **22**, **88** has a curved portion towards the proximal end of the plate **12**, **14**, **16** and a straight portion towards the distal end of the guide bar **11**.

[0102] Forward of the openings **22**, **88**, at least one of the plates **12**, **14**, **16** may be configured to define a keyhole shaped opening **26**, **52**, **90**. As illustrated in the figures, the first plate **12** may define a keyhole shaped opening **26** and the second and third plates **14**, **16** define openings **52**, **90** that cooperate with the keyhole shaped opening **26** of the first plate **12**. While the figures only show the first plate **12** with a keyhole shaped opening **26**, it is contemplated that one, two, or all three of the plates **12**, **14**, **16** may define a keyhole shaped opening **26**, **52**, **90**. For example, it is contemplated that the third plate **16** may define a keyhole shaped opening **52**, and the first and second plates **12**, **14** may define general openings **26**, **90**, such as openings in the shape of a square, circle, oval, or the like. Referring to FIG. **9**, the keyhole shaped opening **26** is shaped so the widest width portion of the opening **26** can receive a fastening feature of a mount for the saw blade cartridge **10**. The openings **26**, **52**, **90** defined in the respective plates **12**, **14**, **16** may be centered between the opposed edges of the guide bar **11**, such that the openings **26**, **52**, **90** are positioned on the longitudinal axis L extending between the proximal end **18** and the distal end **20** of the guide bar **11** such that the longitudinal axis L bisects the guide bar **11**. The keyhole shaped opening **26** may further be formed so the narrower portion of the opening **26** is located distally forward of the wide width portion of the opening **26** to allow the fastening feature **305** of the mount **303** to slide within the opening **26**.

[0103] The guide bar **11** may further comprise additional openings **28**, **54**, **92**, formed in the first, second, and/or third plates **12**, **14**, **16** distal of the keyhole shaped openings **26**, **52**, **90** configured for alignment and/or mounting of the saw blade cartridge. For example, each of the first, second, and/or third plates **12**, **14**, **16** may define an opening **28**, **54**, **92** centered on the longitudinal axis L of the guide bar **11**. As illustrated in the figures, the opening **28**, **54**, **92** may be positioned distal of the keyhole shaped openings **26**, **52**, **90** on the guide bar **11**. However, it is also contemplated that the opening **28**, **54**, **92** may be positioned proximally of the keyhole shaped openings **26**, **52**, **90** on the guide bar **11**. The opening **28**, **54**, **92** may be configured for alignment and/or mounting of the saw blade cartridge **10** to a surgical manipulator **301**, such as a robot or handpiece. The opening **28**, **54**, **92** in each of the first, second, and/or third plates **12**, **14**, **16** may be oriented and/or positioned on the respective plate **12**, **14**, **16** such that the opening are generally aligned and or concentric with one another when the first, second, and/or third plates **12**, **14**, **16** are layered upon one another to define a through-hole through the guide bar **11**. The opening **28**, **54**, **92** may be formed as a circle, oval, square, rectangle, or other similar shape. Furthermore, it is also contemplated that each of the respective openings **28**, **54**, **92** may define a different shape in the respective plates **12**, **14**, **16** and/or define openings of varying sizes. Only one of the openings **28**, **54**, **92** may be required for the purpose of alignment of the saw blade cartridge **10** within the surgical manipulator **301**. Therefore, only one of the openings **28**, **54**, **92** may need to be sized and/or shaped to engage an alignment feature **309** of the mount **303** (explained in greater detail below). The remaining openings **28**, **54**, **92** that do not engage the alignment feature **309** may then be sized and/or shaped such the openings **28**, **54**, **92** do not contact the alignment feature **309** and/or do not interfere with the opening(s) **28**, **54**, **92** that is sized and/or shaped to engage the alignment feature **309**. For example, as illustrated in FIG. **1**, it can be seen that the combination of the openings **28**, **54**, **92** in the respective plates **12**, **14**, **16** define a through-hole through the guide bar **11**. However, it can also be seen that the opening **54** in the third plate **16** is defined to be slightly smaller than the openings

28, **92** of the first and second plates **12**, **14** (only the opening **54** of the third plate **16** can be seen as being smaller than the opening **92** of the second plate **92** in FIG. **1**). In this exemplary configuration of the guide bar **11**, the opening **54** of the third plate **16** is configured to engage the alignment feature **309** of the mount **303** for the purpose of orienting the saw blade cartridge **10** relative to the surgical manipulator **301**, and the opening(s) **28**, **92** of the first and second plates **12**, **14** are sized and shaped such that they do not contact the alignment feature **309** and/or do not interfere with the opening **54** of the third plates **16** ability to engage the alignment feature **309**. As mentioned above, while the opening **54** of the third plate is illustrated as being sized and/or shaped to engage the alignment feature **309** of the mount **303**, any of the opening(s) **28**, **54**, **92** may be sized and/or shaped to engage the alignment feature **309**.

[0104] The guide bar **11** may further comprise additional openings **30**, **94** formed in the first, and/or second plates **12**, **14**, optionally distal of the keyhole shaped openings **26**, **52**, **90**, and configured for verification and/or registration of the saw blade cartridge **10** with a surgical navigation system. For example, at least one of the first and/or second plates **12**, **14** may define an opening **30**, **94**, shown centered on the longitudinal axis L of the guide bar **11**, and oriented and/or positioned to be aligned when the first and second plates **12**, **14** are layered upon one another. The first and/or second plates **12** may comprise openings **30**, **94** configured to provide access to the reference feature **56** defined in the third plate **16**. The reference feature **56** may comprise an aperture defined in the third plate **16** configured as a known contact point for verification and/or registration of the pose of the saw blade cartridge with the surgical navigation system. The reference feature **56** may comprise an aperture in the shape of a triangle, such that the three edges of the aperture defining the triangle act to center a probe or similar device used to touch-off on the reference feature **56**. It is contemplated that a reference feature **56** may comprise an aperture in the shape of a circle, square, rectangle, or other similar polygon. Alternatively, the reference feature **56** may comprise a divot defined in the surface of the third plate **16**. This may include a divot on a first surface of the third plate that is accessible through the opening **30** in the first plate **12** and/or a divot on a second surface of the third plate that is accessible through the opening **94** defined in the second plate **14**. It is further contemplated that the reference feature **56** may comprising a marking. The marking may be realized as an etch, color, engraving, or similar optical marking. The reference feature **56** may be centered on the longitudinal axis L of the guide bar **11**.

[0105] As described above, the proximal portion **18** of the third plate **16** comprises a number of mounting features and/or contact points intended to facilitate alignment and/or mounting of the saw blade cartridge **10** with the surgical manipulator. Therefore, having the reference feature **56** defined and/or disposed on the third plate **16** may provide the advantage of improved accuracy in verification and/or registration of the saw blade cartridge **10** with the surgical navigation system. Furthermore, it is contemplated that the saw blade cartridge **10** may comprise a plurality of reference features **56**. See FIGS. **33** and **34**, which will be described in greater detail below.

[0106] It should be appreciated that the dimension of the reference feature **56** has a predetermined dimensional relationship with respect to the one or more mounting features. This enables the aforementioned registration and/or verification with the surgical navigation system. The predetermined dimensional relationship may be stored on a memory unit of the surgical navigation system and compared to the measured relationship as determined by a tracked pointer having its distal end positioned at the reference feature. By confirming that the reference feature **56** in the localizer coordinate system is at the expected position, the registration verification step can be performed.

[0107] The third plate **16**, as seen best in FIGS. **8** and **9**, is formed to have a proximal portion **18**B of the plate. A pair of laterally spaced apart tines **46**A, **46**B extend forward from the proximal portion **18**B of the third plate **16**. The tines **46**A, **46**B may be referred to as the outer tines. The outer surfaces of tines **46**A, **46**B are the outer surfaces of the third plate **16** that form sections of the outer side surfaces of the guide bar **11**. The outer tines **46**A, **46**B thus have proximal sections that extending forward from proximal portion **18**B and taper first outwardly and then inwardly as you move distally along the third plate **16**. The outer tines **46**A, **46**B are symmetrically located relative to the longitudinal axis L of the guide bar **11**. Outer tines **46**A, **46**B are further shaped to be mirror images each other, mirrored about the longitudinal axis L. Furthermore, the outer tines **46**A, **46**B may be configured to couple the couple the first, second and third plates **12**, **14**, **16** together to form the guide bar **11**. For example, the first and second plates **12**, **14** may be welded or similarly coupled to the outer tines **46**A, **46**B of the third plate **16** to form the guide bar **11**. Alternatively, it is also contemplated that the first and second plates **12**, **14** may be welded or similarly coupled to the inner tine **48** of the third plate **16** to form the guide bar **11**. By welding or similarly coupling the first and second plates **12**, **14** to the inner tine **48** of the third plate **16**, it may avoid any interference, unintended warping or similar disturbance to the edges of the proximal portion **18** of the third plate **16**, which may utilized in aligning and/or orienting the guide bar **11** relative to the surgical manipulator.

[0108] Adjacent the distal end of each tine **46**A, **46**B, is shaped to have a lobe **60**A, **60**B, that extends inwardly towards the longitudinal axis L of the guide bar **11**. Moving proximally to distally along the tine **46**A, **46**B, the associated lobe **60**A, **60**B first curves inwardly toward the longitudinal axis L of the guide bar **11** and then curves outwardly.

[0109] The third plate **16** may further comprise a third tine **48**, the inner tine, that is disposed between the outer tines **46**A, **46**B and is centered on the longitudinal axis L of the guide bar **11**. As described above, the inner tine **48** may be formed to define an opening **52** which may be a keyhole shaped opening **52**. The inner tine **48** may shaped so that opening **52** is identical in shape and positioned to be in registration with opening **26** and **90** of first and second plates **12**, **14**, respectively. The inner tine **48** is of constant width except for the distal end of the tine. At the most distal end, the inner tine **48** may comprise a head **58** with a distally directed face that is curved and configured to define a pivot surface. While not illustrated in the figures, it is also contemplated that the head **58** may be defined a recessed surface configured to receive a rocker or curved surface that may pivot within the recessed surfaced defined by the head **58**. The head **58** may include a center, the center defining a radius of the distally directed face that is curved to define the pivot surface. The center, similar to the opening **54** and the reference feature **56** of the third plate **16**, may be positioned on the longitudinal axis L of the guide bar **11**. This creates alignment of the head **58**, opening **54**, and the reference feature **56** along the longitudinal axis L of the guide bar **11**, and may aid in alignment and/or registration of the blade cartridge **10** with the navigation system and or positioning of the blade cartridge **10** relative to the surgical manipulator. It can be advantageous to have the features for mounting and/or aligning the cartridge relative to the surgical manipulator, such as the opening **54** and/or keyhole shaped opening **52** on the same axis as the head **58** about which the blade is manipulated and the reference feature **56**. This can lead to better alignment of the blade cartridge **10** with the surgical manipulator, which can also then lead to improved tolerances with regard to the registration of the blade cartridge **10** to the navigations system. Furthermore, in the provided exemplary configuration of the blade cartridge **10**, the opposed edges of the proximal portion **18**B of the third plate **16** may be configured as a mounting feature for the saw blade cartridge **10**. In the exemplary configuration, all of the various mounting features, alignment features, and navigation registration features may by located and/or disposed on the third plate **16**. This can result in improved tolerances and/or repeatability of positioning the blade cartridge **10** relative to the surgical manipulator, resulting in more accurate position identification and by extension, navigation of the blade cartridge **10** during a medical procedure.

[0110] While the figures illustrate an inner tine disposed between the outer tines **46**A, **46**B, it is contemplated that the third plate **16** may be formed without the inner tine **48**, and only include the two outer tines **46**A, **46**B spaced apart from one another. In such a configuration, a pivot pin may be coupled adjacent the blade head. In other words, while the third plate is shown has having three tines formed as a single component, it is contemplated that the three tines may be formed separately and secured between the first and second plates.

[0111] The outer tines **46**A, **46**B are spaced apart from the inner tine **48**, such that the third plate **16** defines a pair of elongated, closed end slot **50**A, **50**B between each of the outer tines **46**A, **46**B and the inner tine **48**. The proximal end of the slot **50**A, **50**B is the closed end of the slot. When cartridge **10** is assembled, each slot **50**A, **50**B is disposed over a separate one of the openings **22**, **88** in the first and second plates **12**, **14**, respectively. The third plate **16** is further formed so that the proximal end portion of each slot **50**A, **50**B has a shape substantially equal to the shape of openings **22**, **88** in the first and second plates **12**, **14**. Thus, the lateral width of the proximal end portion of each slot **50**A, **50**B is greater than the width of the slot **50**A, **50**B distally forward of the proximal portion **18**.

[0112] While the third plate **16** is generally depicted in the figures as being formed as a single solitary or unitary plate, it is contemplated that the third plate **16** may be configured as a plurality of separate pieces arranged to form the features and/or portions of the third plate **16** that are described above. For example, the proximal portion **18** of the third plate **16** may be formed as a unitary piece with the outer tines **46**A, **46**B, and the inner tine **48** may be a separate piece. Alternatively, the inner tine **48** may be omitted. In yet another exemplary configuration, the proximal portion **18** of the third plate **16**, the outer tines **46**A, **46**B, and the inner tine **48** may all be formed as separate pieces.

[0113] Referring to FIG. **9**, it is observed that first plate **12** is shaped to have an outer perimeter substantially identical to that of second plate **14**. However, it is also contemplated that the one of the firs plate **12** or the second plate **14** may be longer/shorter and/or narrower/wider than the other plate. For example, the second plate **14** may be constructed such that the outer perimeter is narrower, i.e. the perimeter edge is closer to the longitudinal axis L, than that of the first plate **12**. Forward of the proximal portion **18** of the first and second plates **12**, **14** is a distal portion **20**. The distal portion of the first and second plates **12**, **14** is shaped to have two openings **24**, **96**. The openings **24**, **96** are identical in shape and positioned such that the openings **24** of the first plate **12** are in registration with the openings **96** of the second plate **96**.

[0114] The saw blade cartridge **10** may further comprise a saw blade **70** disposed between the first and second plates **12**, **14** at the distal end of the saw guide bar **11**. The saw blade **70** is formed to have a blade body **72** and a blade head **74**. The saw blade **70** is formed so that blade body has a thickness that is no greater than the thickness of third plate **16**. The blade body **72** of the saw blade **70** seen in FIGS. **8** and **9** is generally rectangular in shape. However, it is also contemplated that the perimeter edges **73** of the blade body **72** may be tapered and/or curved. The taper and/or curve may be matched to the taper or curve of the lobes **60**A, **60**B of the outer tines **46**A, **46**B of the inner plate **16**. The shape of the blade body may be configured to allow the blade body **72** to pivot about the head **58** of the inner tine **48**, oscillating between the lobes **60**A, **60**B of the outer tines **46**A, **46**B.

[0115] When saw blade **70** is in the centered position within the guide bar **11** the major axis of the blade body **72** is collinear with the longitudinal axis L of the guide bar **11**. The blade body **72** may be shaped to have two feet **84**, one foot identified. The feet **84** have the same thickness of as the blade body **72**. Each foot **84** extends to the open end of the adjacent slot **50**A, **50**B internal to the inner tine **48**. Between the feet **84**, the blade body **72** is formed to have a curved, proximally directed face **80**. The blade body **72** is shaped so that face **80** can seat against and pivot around the head **58** integral with the inner tine **48**. While the figures illustrate the third plate with an inner tine **48** defining a head **58** for the saw blade to pivot about, it is further contemplated that the guide bar may be configured with a pin disposed between the first and second plates **12**, **14**, and the saw blade **70** being configured to pivot about the pin. The blade feet **84** and blade body **72** are sometimes referred to as the base of the saw blade **70**.

[0116] The distal end of the blade body **72** extends distally of the guide bar **11**, such that the blade head **74** extends forward of the distal end of guide bar **11**. The blade head **74** is formed with a plurality of teeth **76**, **78**. The saw blade **70** is further formed so that blade head **74** has a thickness greater than that of the blade body **72**. More particularly, the blade head **74** is formed to have a thickness equal to or greater than the thickness of the guide bar **11** so that the kerf formed by the cutting action of saw blade **70** is sufficient to receive the guide bar **11**.

[0117] Referring to FIGS. **2** through **5**, the blade head **74** may comprise two type of teeth, base teeth **78** and one or more pilot teeth **76**. For example, as illustrated in FIG. **2**, the blade head **74** may comprise two pilot teeth **76**A, **76**B. However, it is contemplated that the blade head **74** may comprise a single pilot tooth, or more than two pilot teeth. The pilot teeth **76**A, **76**B may be disposed proximate the center of the distal surface of the blade head **74**. Furthermore, the pilot teeth **76**A, **76**B may be configured to extend a first distance DI distally beyond the adjacent base teeth **78** of the blade head **74**, as best seen in FIG. **5**. In operation, with the pilot teeth **76**A, **76**B extending distally beyond the adjacent base teeth **78**, the pilot teeth **76**A, **76**B are configured to oscillate along a first arc distance **77** as the adjacent base teeth **78** oscillate about a second arc distance **75**, such that the first arc distance **77** is distal to the second arc distance **75**.

[0118] Referring to FIGS. **5** through **7**, the saw blade **70** may comprise a central axis C that extends from the proximal to the distal end of the saw blade **70**. A first plane, Plane-**1**, may be oriented to intersect the central axis C of the saw blade **70** and bisect the saw blade into left and right halves. A second plane, Plane-**2**, may be oriented to intersect the central axis C of the saw blade **70** and bisect the saw blade into top and bottom halves. Plane-**1** and Plane-**2** may be oriented to be perpendicular to one another. The pilot teeth **76**A, **76**B may be configured to extend distally from the blade head **74** and positioned such that they are on opposing sided of Plane-**1**.

[0119] A third plane, Plane-**3**, may be configured to intersect the central axis C of the saw blade **70**. Plane-**3** may be oriented at a mirror angle, θ, relative to either of Plane-**1** and/or Plane-**2**. For example, Plane-**3** is oriented at a mirror angle, θ, that is defined relative to Plane-**2**. The mirror angle θ, as measured relative to plane-**2**, may be any angle ranging from 20 to 80 degrees. For example, plane-**3** may be oriented at a mirror angle θ of 75 degrees relative to plane-**2**. The Plane-**3** may also be referred to as the mirror-plane, as the pilot teeth **76**A, **76**B may be mirrored across plane-**3**. For example. the second pilot tooth **76**B may be oriented such that it is a mirrored representation of the first pilot tooth **76**A mirrored about plane-**3**, the mirroring plane.

[0120] It is also contemplated that each of that each of the pilot teeth **76**A, **76**B may be oriented in a cross-cut pattern, wherein adjacent teeth are oriented at opposing angles relative to one another. For example, referring back to FIGS. **3** and **4**, a side profile of the blade head **74** is illustrated illustrating that the pilot teeth **76**A, **76**B are oriented in a cross-cut pattern. The first pilot tooth **76**A is oriented in a generally downward direction relative to Plane-**2**, and the second pilot tooth **76**B that is positioned adjacent the first pilot tooth **76**A is oriented in a generally upward direction relative to Plane-**2**, with the pilot teeth **76**A, **76**B defining a v-shaped side at the distal end of the blade head **74**.

[0121] A drive link **62**A, **62**B is disposed in each of the slots **50**A, **50**B internal to the guide bar **11**. Each drive link **62**A, **62**B is in the form of an elongated flat strip of metal. The drive links **62**A, **62**B are formed so that, at the proximal end of each link, there is a foot **64**. Each foot **64** is formed to have a center located through hole **66**, one opening identified. Through holes **66** are dimensioned so that the associated drive link feet **64** can be fitted over the saw head drive pins. Each drive link **62**A, **62**B is shaped so that foot **64** has a thickness that is greater than the thickness of the metal strip forming the main body of the link **62**A, **62**B. The thickness of the feet **64** is typically no greater than the thickness of the guide bar **11**. When the cartridge **10** is assembled, the feet **64** generally seat in the proximal end portion of the associated slot **50**A, **50**B. The portions of the feet **64** that project outwardly from the main body of each link **62**A, **62**B seat in the openings **22**, **88** of the first and/or second plates **12**, **14**.

[0122] The drive links **62**A, **62**B may further comprise two fingers **68** extend distally forward from the distal end of the main body of each drive link **62**A, **62**B, one finger identified. Fingers **68**

overlap and are spaced apart from each other. More particularly, fingers **68** are spaced apart from each a sufficient distance so that a blade foot **84** can seat between each pair of fingers. Each finger **68** is formed with a hole **82**. The holes **82** of each pair of fingers **68** are in registration with each other.
[0123] As part of the process of assembly a cartridge **10**, the saw blade **70** is positioned so that each blade foot **84** is disposed between a pair of drive link fingers **68**. A pivot pin may then extend through the finger holes **84** and an opening blade foot to pivotally holds the foot **84** to the associated drive link **62**A, **62**B.
[0124] During the assembly of the cartridge **10**, the third plate **16** is initially welded or otherwise secured to the first plate **12** and/or to the second plate **14**. After this operation is completed, the drive links-and-blade assembly is positioned so that the drive links **62**A, **62**B are seated in slots **50**A, **50**B and the curved proximally directed face **80** of blade **70** is seated against the curved distally directed face of head **58** integral with the third plate **16**. The second plate **14** and/or first plate **12** is then welded or otherwise secured to the exposed face of the third plate **16**. At the completion of the process of assembling the cartridge **10**, the drive link feet **64** seat in openings **22**, **88** of the first and second plates **12**, **14**. The drive link fingers **68** seat in openings **24**, **96** formed, respectively in the first and second plates **12**, **14**. Blade head **74** is located immediately forward of the distal end of the guide bar **11**.
[0125] Referring to FIGS. **10** and **11**, the distal portion of the guide bar **11** may comprises a pair of generally oval shaped opening **98** in the second plate **14**. While two openings **98** are illustrated in the figures, it is contemplated that guide bar may be constructed with only a single opening, or with more than two openings. Furthermore, while the openings are illustrated in the figures as being defined in the second plate **14**, it is contemplated that the similar openings may also be defined in the first plate **12**. The openings **98** may comprise a curve and/or bend that extends longitudinally along the length of the distal portion **20** of the second plate **14**. The opening may be configured to assist with the removal of debris that may become trapped between the first and second plates **12**, **14** when the saw blade cartridge **10** is utilized to cut biological tissue.
[0126] The distal portion **20** of the guide bar **11** may also be formed to define a cut-out or slot **38** between the first and second plates **12**, **14**. As illustrated in FIGS. **10** and **11**, the outer tine **46** of the inner plate **16** is formed so that the distal most point of the lobe **60** of the outer tine **46** is positioned proximal of the distal most edge of the first and second plates **12**, **14**. This arrangement defines a slot **38** along the side edge of the guide bar **11** that may assist the removal of debris that may become trapped between the first and second plates **12**, **14** when the saw blade cartridge **10** is utilized to cut biological tissue. The slot **38** along the side edge of the guide bar **11** that may also assist to increase the range of motion of the saw blade **70** as it is pivoted. By shortening the length of the outer tines **46**, increased clearance may be provided allowing the saw blade to pivot further, and thus increasing the distance the blade head **74** may oscillate.
[0127] While not illustrated in the figures, it is further contemplated that the saw blade cartridge **10** may include ribs on the outer surface of the guide bar. The ribs may be aligned with the longitudinal axis L of the guide bar **11**. The ribs may be formed separate from the first and/or second plates **12**, **14**, and welded or otherwise permanently secured to the rest of cartridge **10**. The ribs may reduce the flexing of the guide bar **11**. An exemplary configuration of a saw blade cartridge is disclosed in PCT Pub. No. WO 2013/016472A1/US Pat. Pub. No. 2014/0163558, which is explicitly incorporated herein by reference.
[0128] Referring to FIGS. **12** to **15**, a second configuration of the saw blade cartridge **110** is illustrated. It should be understood that components and/or features of the second configuration of the saw blade cartridge **110** including the same base reference number increased by one hundred, i.e. **10** and **110**, may operate in a similar or the exact same manner as described above.
[0129] Similar to as described above, the second configuration of the saw blade cartridge **110** may comprises a guide bar **111** including a plurality of plates (**112**, **114**, **116**), including a first plate **112**,

a second plate **114**, and a third plate **116**. The guide bar **111** may be configured such that the third plate **116** may be disposed between the first plate **112** and the second **114**. The first, second, and third plates, **112**, **114**, **116** are constructed in the same or similar fashion as those of the saw blade cartridge **10** described above. However, the configuration of the saw blade cartridge **110** illustrated in FIGS. **12** to **15** is constructed with an alternative configuration of a saw blade **170**.

[0130] Referring to FIG. **13**, the saw blade **170** may comprise a blade body **172** that extends distally of the guide bar **111**, such that a blade head **174** coupled to the distal end of the blade body **172** extends forward of the distal end of guide bar **111**. The blade head **174** is formed with a plurality of teeth **176**, **178**. The saw blade **170** is further formed so that blade head **174** has a thickness greater than that of the blade body **172**. More particularly, the blade head **174** is formed to have a thickness equal to or greater than the thickness of the guide bar **111** so that the kerf formed by the cutting action of saw blade **170** is sufficient to receive the guide bar **111**.

[0131] The blade head **174** may comprise two type of teeth, base teeth **178** and one or more pilot teeth **176**. For example, as illustrated in FIG. **13**, the blade head **74** may comprises a single pilot tooth **176**. As described above, the blade head **174** may also comprise two or more pilot teeth. The pilot tooth **176** may be disposed proximate the center of the distal surface of the blade head **174**. Furthermore, the pilot tooth **176** may be configured to extend a second distance D**2** distally beyond the adjacent base teeth **178**A, **178**B of the blade head **174**, as best seen in FIG. **13**. In operation, with the pilot tooth **176** extends distally beyond the adjacent base teeth **178**A, **178**B, the pilot tooth **176** is configured to oscillate along a first arc **177** as the adjacent base teeth **78** oscillate about a second arc distance **175**, such that the first arc distance **177** is distal to the second arc distance **175**.

[0132] Referring to FIGS. **14** and **15**, the saw blade **170** may comprise a central axis C that extends from the proximal to the distal end of the saw blade **170**. A first plane, Plane-**1**, may be oriented to intersect the central axis C of the saw blade **170** and bisect the saw blade into left and right halves. A second plane, Plane-**2**, may be oriented to intersect the central axis C of the saw blade **170** and bisect the saw blade into top and bottom halves. Plane-**1** and Plane-**2** may be oriented to be perpendicular to one another. The pilot tooth **76** may be configured to extend distally from the blade head **174** and on one side of Plane-**1**, with an adjacent base tooth **178**B disposed on the opposing side of Plane-**1**.

[0133] A third plane, Plane-**3**, may be configured to intersect the central axis C of the saw blade **170**. Plane-**3** may be oriented at a mirror angle, θ, relative to either of Plane-**1** and/or Plane-**2**. For example, Plane-**3** is oriented at a mirror angle, θ, that is defined relative to Plane-**2**. The mirror angle θ, as measured relative to plane-**2**, may be any angle ranging from 20 to 80 degrees. For example, plane-**3** may be oriented at a mirror angle θ of 75 degrees relative to plane-**2**. The plane-**3** may also be referred to as the mirror-plane, as the pilot tooth **176** and the adjacent, shorter base tooth **178**B each have surfaces that define a cutting edge that is mirrored across plane-**3**. For example. the adjacent, shorter base tooth **178**B may be oriented such that it has a face that is a mirrored representation of the pilot tooth **176** mirrored about plane-**3**, the mirroring plane.

[0134] It is also contemplated that each of that each of the pilot tooth **176** and adjacent base teeth **178**A, **178**B may be oriented in a cross-cut pattern, wherein adjacent teeth are oriented at opposing angles relative to one another. For example, referring back to FIG. **15**, a side profile of the blade head **174** is illustrated illustrating that the pilot tooth **176** and adjacent base teeth **178**A, **178**B are oriented in a cross-cut pattern. The first pilot tooth **176** is oriented to point in a generally downward direction relative to Plane-**2**, and adjacent base teeth **178**A, **178**B are each oriented in a generally upward direction relative to Plane-**2**. The combination of the pilot tooth **176** and adjacent base teeth **178**A, **178**B being arranged in this configuration results in the side of the blade head **174** (i.e. top or bottom) that the pilot tooth **176** is oriented toward having a leading edge that is distal to the edge defined by the adjacent base teeth **178**A, **178**B. This particular shape of leading edge may be utilized to reduce skiving or diving when approaching a cutting surface at an angle. For example, it may be advantageous to orient the saw blade cartridge **110** relative to the cutting surface, such as a bone or other biological tissue, such the forward most leading edge defined by the pilot tooth **176** contact the cutting surface first.

[0135] Referring to FIGS. **16** to **24**, a third configuration of the saw blade cartridge **210** is illustrated. It should be understood that components and/or features of the second configuration of the saw blade cartridge **110** including the same base reference number increased by one hundred, i.e. **10**, **110**, **210**, may operate in a similar or the exact same manner as described above.

[0136] Similar to as described above, the third configuration of the saw blade cartridge **210** may comprises a guide bar **211** including a plurality of plates (**212**, **214**, **216**), including a first plate **212**, a second plate **214**, and a third plate **216**. The guide bar **211** may be configured such that the third plate **216** may be disposed between the first plate **212** and the second **214**. The first, second, and third plates, **212**, **214**, **216** are constructed in a generally similar fashion as those of the saw blade cartridges **10**, **110** described above. However, the configuration of the saw blade cartridge **210** illustrated in FIGS. **16** to **24** is constructed with an alternative configuration of the first, second, and third plates, **212**, **214**, **216**.

[0137] The saw blade cartridge **210** may comprise a saw blade **270** that is the same or similar to either of the saw blades **70**, **170** described above. For exemplary purposes, the saw blade cartridge **210**, as illustrated in FIG. **16** comprises a saw blade **270** similar to the second configuration of the saw blade **170** described above. The saw blade includes a blade body **272** and a blade head **274**. The blade head **274** may comprise two type of teeth, base teeth **278** and one or more pilot teeth **276**. For example, as illustrated in FIGS. **18** and **19**, the blade head **274** may comprises a single pilot tooth **276**. However, as described above, it is contemplated that the blade head **274** may also comprise two or more pilot teeth. The pilot tooth **276** may be disposed proximate the center of the distal surface of the blade head **274**. Furthermore, the pilot tooth **276** may be configured to extend a third distance D**3** distally beyond the adjacent base teeth **278**A, **278**B of the blade head **274**, as best seen in FIGS. **19** and **20**. In operation, with the pilot tooth **276** extends distally beyond the adjacent base teeth **278**A, **278**B, the pilot tooth **276** is configured to oscillate along a first arc distance **277** as the adjacent base teeth **278** oscillate about a second arc distance **275**, such that the first arc distance **277** is distal to the second arc distance **275**.

[0138] Referring to FIG. **19**, a guide bar **211** including an alternative configuration and/or shape of the distal portion **220** is illustrated. As described above, with regard to FIG. **1**, the opposing edges of the distal portion **20** of the guide bar **11** may comprise a tapered edge **32**, **99** defined by the first and second plates **11**. As illustrated in FIG. **19**, the distal portion **220** of the guide bar **211** may also be formed such that opposing edges comprise both a tapered portion **232**, **299** and a parallel portion **231**, **297**. For example, the distal portion **220** of the guide bar **211** comprise a tapered edge **232**, **299** immediately forward of the middle portion of the guide bar **211**. The opposing edges **232**, **299** of the first and second plates **212**, **214** may be tapered inward toward the longitudinal axis L of the guide bar **211** as you move distally along the guide bar **211**. Forward of the tapered edge **231**, **299**, the guide bar **211** may be formed such that the opposing edges **231**, **297** of the first and second plates **212**, **214** are generally parallel to the longitudinal axis L of the guide bar **211**.

[0139] The distal portion of the guide bar **11** may also comprises a pair of generally oval shaped opening **298** in the second plate **214**. While two openings **298** are illustrated in the figures, it is contemplated that guide bar may be constructed with only a single opening, or with more than two openings. Furthermore, while the openings are illustrated in the figures as being defined in the second plate **214**, it is contemplated that the similar openings may also be defined in the first plate **212**. The openings **298** may comprise a curve and/or bend that extends longitudinally along the length of the distal portion **220** of the second plate **214**. The opening may be configured to assist with the removal of debris that may become trapped between the first and second plates **212**, **214** when the saw blade cartridge **210** is utilized to cut biological tissue.

[0140] Referring to FIG. **23**, an alternative configuration of the outer tines **246**A, **246**B is illustrated. Adjacent the distal end of each tine **246**A, **246**B, each tine is shaped to have a tab **261** one identified, that extends inwardly to the longitudinal axis L of the guide bar **211**. Proximal to the tab **261**, each tine **246**A, **246**B is shaped to have a lobe **260**A, **260**B, that extends inwardly towards the longitudinal axis L of the guide bar **211**. Moving proximally to distally along the tine **246**A, **246**B, the associated lobe **260**A, **260**B first curves inwardly toward the longitudinal axis L of the guide bar **211** and then curves outwardly. The third plate **216** is formed so that lobes **260**A, **260**B are spaced proximally away from tabs **261**.

[0141] Referring to FIGS. **25** to **27**, an exemplary configuration of a mount **303** for securing a saw blade cartridge **10**, **110**, **210** to a surgical manipulator **301** is illustrated. The mount **303** may comprise a coupling rod **305** configured to removably secure the saw blade cartridge **10**, **110**, **210** to the mount **303**. The coupling rod **305** may interact with the openings **26**, **52**, **90**, **126**, **152**, **190**, **226**, **252**, **290** of the saw blade cartridge **10**, **110**, **210**, such as the key hole opening **26**, **126**, **226** to removably couple the saw blade cartridge **10**, **110**, **210** to the mount **303**. The coupling rod **305** may comprise a head configured to fit thought the larger portion of the keyhole opening **26**, **126**, **226**, and the shaft of the coupling rod **305** may be configured to fit within the smaller portion of the keyhole opening **26**, **126**, **226** after the head has been inserted through the larger portion. The larger portion may then be seated above the smaller portion and configured to removably secure the saw blade cartridge **10**, **110**, **210** to the mount **303**. An exemplary configuration of a mount for a saw blade cartridge is disclosed in International Publication No. WO2007/030793 and U.S. Pat. No. 8,696,673, both of which are incorporated herein in their entirety.

[0142] The coupling rod **305** may be controlled by a coupling lever **307** configured to manipulate the coupling rod **305** between a locked and unlocked state. For example, the coupling lever **307** may be rotated in one direction to retract the coupling rod **305** to a locked stated configured to secure the saw blade cartridge **10**, **110**, **210** to the mount **303**. Alternatively, the coupling lever **307** may be rotated in the opposite direction to extend or advance the coupling rod **305** to an unlocked stated configured to allow for installation and/or removal of the saw blade cartridge **10**, **110**, **210** from the mount **303**.

[0143] The mount may further comprise an alignment feature **309** configured to interact with the opening **28**, **54**, **92**, **128**, **154**, **192**, **228**, **254**, **292** in each of the first, second, and/or third plates **12**, **14**, **16**, **112**, **114**, **116**, **212**, **214**, **216**. The alignment feature **309** may comprise a pin configured to be inserted through the opening **28**, **54**, **92**, **128**, **154**, **192**, **228**, **254**, **292** in each of the first, second, and/or third plates **12**, **14**, **16**, **112**, **114**, **116**, **212**, **214**, **216** when the saw blade cartridge **10**, **110**, **210** is coupled to the mount **303**. alignment feature **303** may be sized to simultaneously engage opposing lateral surfaces of the first opening **28**, **54**, **92**, **128**, **154**, **192**, **228**, **254**, **292**. The alignment feature **309** may be defined to have a round, oval, square, diamond, or similar polygonal shape. While the shape is not critical, in configurations where the alignment feature **309** is intended to provide the functionality of aligning the saw blade cartridge **10** with the surgical manipulator **301**, the width of the surgical alignment feature **309** (as measured in a direction generally perpendicular to the lateral axis L of the saw blade cartridge **10**) may be sized to simultaneously engage opposing lateral surfaces of the first opening **28**, **54**, **92**, **128**, **154**, **192**, **228**, **254**, **292**. The combination of the coupling rod **305** and the alignment feature **309** may be configured to orient the longitudinal axis L of the saw blade cartridge **10**, **110**, **210** with the mount **303**. Proper and precise alignment of the saw blade cartridge **10**, **110**, **210** with the mount can be particularly advantageous when the saw blade cartridge **10**, **110**, **210** and/or the mount **303** do not have their own dedicated tracker. For example, the pose of the saw blade cartridge **10**, **110**, **210** may be determine based on a tracker **4**A, **4**B coupled to the surgical manipulator **301** and/or the mount **303** and a transformation based on known information about the mount **303** and/or the saw blade cartridge **10**, **110**, **210**. In this particular scenario, the alignment of the saw blade cartridge **10**, **110**, **210** with the mount **303** is important to accurately tracking the pose of the saw blade cartridge **10**, **110**, **210**.

[0144] The mount **303** may also comprise a mount surface **315** configured to receive the saw blade cartridge **10**, **110**, **210**. When the saw blade cartridge **10**, **110**, **210** is coupled to the mount, bottom surface of the proximal portion of the first plate **12**, **112**, **212** may seated adjacent the mounting surface **315**. When the coupling rod **305** is in the locked state, the coupling rod **305** may be configured to press the saw blade cartridge **10**, **110**, **210** against the mount surface **315** to hold the saw blade cartridge **10**, **110**, **210** in place.

[0145] A method of the surgical saw blade **10** to the surgical manipulator **301** may include inserting the alignment feature **309** of the surgical manipulator **301** in a first opening **28**, **54**, **92**, **128**, **154**, **192**, **228**, **254**, **292** of the surgical saw blade **10** and inserting the coupling rod **305** of the surgical manipulator **301** in the second opening **26**, **52**, **90**, **126**, **152**, **190**, **226**, **252**, **290** of the surgical saw blade **10**. The method may also includes sliding the surgical saw blade **10** proximally toward the surgical manipulator **301** causing each of the alignment feature **303** and the coupling rod **305** to move from a proximal position within each of the first opening **28**, **54**, **92**, **128**, **154**, **192**, **228**, **254**, **292** and the second opening **26**, **52**, **90**, **126**, **152**, **190**, **226**, **252**, **290** respectively to a distal position within each of the first opening **28**, **54**, **92**, **128**, **154**, **192**, **228**, **254**, **292** and the second opening **26**, **52**, **90**, **126**, **152**, **190**, **226**, **252**, **290**. The alignment feature **309** may be sized and/or shaped to simultaneously engage opposing lateral surfaces of the first opening **28**, **54**, **92**, **128**, **154**, **192**, **228**, **254**, **292** when positioned within the distal position of the first opening **28**, **54**, **92**, **128**, **154**, **192**, **228**, **254**, **292** to axially align the surgical saw blade **10** relative to the surgical manipulator **301** as the surgical saw blade **10** is slid proximally toward the surgical manipulator **301**. It is further contemplated that the alignment feature **309** may be sized and/or shaped to engage the first opening **28**, **54**, **92**, **128**, **154**, **192**, **228**, **254**, **292** to transversely align the surgical saw blade **10** relative to the surgical manipulator **301**.

[0146] The mount **303** may also comprise a drive rod **311** for actuating the saw blade **70**, **170**, **270** of the saw blade cartridge **10**, **110**, **210**. The drive rod **311** may comprise at least one drive pin **313** configured to be seated in the holes **66**, **166**, **266** formed in a drive link foot **64**, **164**, **264**. While a pair of drive pins are illustrated in the figures, it is contemplated that the mount **303** may be configured to actuate the saw blade **70**, **170**, **270** of the saw blade cartridge **10**, **110**, **210** with only a single drive pin **313**. In operation, the drive rod **311** may be oscillated and/or partially rotated back and forth causing the drive pin(s) to reciprocate, and in turn reciprocating the drive link **62**, **162**, **262**.

[0147] The drive rod **311** may also comprise a biasing member (not shown) configured to urge the drive rod **311** in a proximal direction toward a locked position. The force of biasing member may be overcome by pulling the drive rod **311** in a distal direction toward a loading position. In operation, the holes **66**, **166**, **266** formed in a drive link foot **64**, **164**, **264** may be seated over the drive pin(s) **313** while the drive rod **311** is in the locked position. The saw blade cartridge **10**, **110**, **210** may then be pulled distally (forward) to overcome the force of the biasing member and urging the drive rod toward the loading position where the saw blade cartridge **10**, **110**, **210** can be mounted to the mount **303**. Once the saw blade cartridge **10**, **110**, **210** is loaded, i.e. the coupling rod is inserted through the key hole opening **26**, **126**, **226** and the proximal portion **18**, **118**, **218**, of the first plate **12**, **112**, **212** is adjacent the mounting surface **315**, the saw blade cartridge **10**, **110**, **210** may be released and the biasing member will move the saw blade cartridge **10**, **110**, **210** proximally and until it has reached the locked position.

[0148] The mount **303** may further comprise a bracket **317** positioned on opposing sides of the mounting surface **315**. The bracket may extend upward from the mounting surface and comprise a tab **319** that is configured to at least partially extend over the top surface of the second plate **14**, **114**, **214** of the saw blade cartridge **10**, **110**, **210** when the saw blade cartridge **10**, **110**, **210** is coupled to the mount **303**. The purpose of the pulling the saw blade cartridge **10**, **110**, **210** distally (forward) to load the saw blade cartridge **10**, **110**, **210** on the mount is to allow the proximal portion **18**, **118**, **218** of the saw blade cartridge **10**, **110**, **210** to clear the tabs **319** of the brackets **317**. Then when the biasing member moves the saw blade cartridge **10**, **110**, **210** proximally, a portion of the proximal portion **18**, **118**, **218** of the saw blade cartridge **10**, **110**, **210** is seated between the tab **319** and the mounting surface **315**.

[0149] The bracket **317** may also comprise a tapered surface **325**. The tapered surface **325** may be configured to match the various tapers of the opposing edges of the proximal portion **18**, **118**, **218** of the third plate **16**, **116**, **216** for the purpose of orient and/or securing the saw blade cartridge **10**, **110**, **210** to the mount **303**. The tapered surface **325** may be configured to match the various tapers of the opposing edges of the proximal portion **18**, **118**, **218** of the third plate **16**, **116**, **216** to such the biasing member urging the saw blade cartridge **10**, **110**, **210** proximally will cause the proximal portion **18**, **118**, **218** of the third plate **16**, **116**, **216** to be wedged between the opposed tapered surface **325** of the bracket **317** to axially align the surgical saw blade **10** relative to the surgical manipulator **301**. It is also contemplated that the third plate **16**, **116**, **216** to be wedged between the opposed tapered surface **325** of the bracket **317** to transversely align the surgical saw blade **10** relative to the surgical manipulator **301**. The opposing edges of the proximal portion **18**, **118**, **218** of the third plate **16**, **116**, **216** formed with various segments having different taper angles and the tapered surface **325** of bracket **317** being formed to match can provide improved alignment of the saw blade cartridge **10**, **110**, **210** with the mount **303**, improving the accuracy of the navigation of the saw blade cartridge **10**, **110**, **210**.

[0150] Referring to FIGS. **28** to **32**, an exemplary configuration of the surgical assembly **1** is illustrated, including a saw blade cartridge **10**, **110**, **210** coupled to a mount **303** of a surgical manipulator **301**.

[0151] In operation, the saw blade cartridge **10**, **110**, **210** may be coupled to a surgical manipulator **301**, such as a robot arm, or a hand-hand power tool, including a hand-held robotic handpiece. The saw blade cartridge **10**, **110**, **210** may be fitted to a mount **303** of the surgical manipulator **301**. To accomplish this, each drive pin **313** of the mount **303** is seated in a separate one of the holes **66**, **166**, **266** formed in a drive link foot **62**, **162**, **262**, and the saw blade cartridge **10**, **110**, **210** may be pulled forward pulling the drive pins **313** and the driver mechanism **311** forward. The saw blade cartridge **10**, **110**, **210** may then be lowered such that the bottom surface of the first plate **12**, **112**, **212** is positioned adjacent the mounting surface **315** of the mount **313**. In doing so, the coupling rod **305** of the mount may be inserted through the overlapping openings **26**, **52**, **90**, **126**, **152**, **190**, **226**, **252**, **290** of the saw blade cartridge **10**, **110**, **210**. The coupling rod **305** may then be lowered using the coupling lever **307** to lower the head of the coupling rod **305** onto the guide bar **11**, **111**, **211** so the head of the coupling rod **305** presses against the top surface of which if the plates **12**, **14**, **16**, **112**, **114**, **116**, **212**, **214**, **216** include the key-hole shaped opening **26**, **126**, **226**. This press action holds the saw blade cartridge **10**, **110**, **210** to the mount **303**. When the saw blade cartridge **10**, **110**, **210** is secured to the mount **303**, the drive pins **313** and drive links **311** cooperate to urge the saw blade cartridge **10**, **110**, **210** proximally. The proximal portion **18**, **118**, **218** of the saw blade cartridge **10**, **110**, **210** is seated adjacent the mount surface **315**.

[0152] The saw blade cartridge **10**, **110**, **210** may be actuated by actuating of the motor (not shown) of the surgical manipulator **301**. The actuation of the motor results in the back-and-forth oscillation of the drive pins **313**. The movement of the drive pins **313** causes the drive links **62**, **162**, **262** to engage in opposed back and forth reciprocation of the drive links **62**, **162**, **262**. The opposed back and forth motion of the drive links **62**, **162**, **262** causes the saw blade **70**, **170**, **270** to pivot back and forth around the head **58**. **158**, **258** of the guide bar **11**, **111**, **211**. The pivoting action of the saw blade **70**, **170**, **270** causes oscillation of the blade head **74**, **174**, **274**, and by extension the blade teeth **76**, **78**, **176**, **178**, **276**, **278** causing the teeth to cut the tissue against which the saw blade cartridge **10**, **110**, **210** is pressed.

[0153] Once the saw blade cartridge **10**, **110**, **210** has been mounted to the mount **303** and/or the surgical manipulator **301**, the reference feature **56**, **156**, **256** of the saw blade cartridge **10**, **110**, **210** may be utilized to register and/or verify the pose of the saw blade cartridge **10**, **110**, **210** with the navigation system. A method of registering the saw blade cartridge **10**, **110**, **210** with a navigation system may comprise touching a registration tool or instrument **2** including a first tracking device **3** to the reference feature **56**, **156**, **256** on the inner plate **16**, **116**, **216** by inserting a distal end of the instrument through the aperture **30**, **94**, **130**, **194**, **230**, **294** in one of the first plate **12**, **112**, **212** or the second plate **14**, **114**, **214**. The reference feature **56 156**, **256** may be configured to register the position and/or orientation of the saw blade cartridge **10**, **110**, **210** to the navigation system, and/or verify the position and/or orientation of the saw blade cartridge **10**, **110**, **210** in the surgical navigation system. The method may further comprise tracking the position and orientation of the first tracking device **3** as the instrument **1** is touched to the reference feature **56**, **156**, **256** on the saw blade cartridge **10**, **110**, **210** and determining an actual position and orientation of the reference feature **56 156**, **256**. The method may also comprise determining an expected position of the reference feature **56 156**, **256** based on predetermined geometrical data corresponding to the relationship between a mounting feature and the reference feature **56**, **156**, **256** of the saw blade cartridge **10**, **110**, **210** as described above.

[0154] The actual position and orientation of the reference feature **56 156**, **256** may further be configured to identify the type of saw blade cartridge **10**, **110**, **210**. For example, the position of the reference feature **56 156**, **256** may vary between different types and/or size blades. Therefore, when the actual position and orientation of the reference feature **56 156**, **256** is determine, the navigation system may be configured to identify the type of saw blade cartridge **10**, **110**, **210** based on the actual position and orientation of the reference feature **56 156**, **256**. This may include identifying a characteristics of the saw blade cartridge **10**, **110**, **210**, such as, but not limited to, a type of the blade, a blade thickness of the blade head, a tooth configuration of the blade head, a length of the guide bar, and/or a width of the guide bar **11**, **111**, **211**.

[0155] Referring to FIGS. **33** to **36** the saw blade cartridge **10** described above is illustrated including an alternative configuration of reference features **56**, **56**A, **56**B, **56**C, **56**D. As described above, the saw blade cartridge **10** may comprise a plurality of reference features **56**, **56**A, **56**B, **56**C, **56**D. FIGS. **33**, **34**, and **36** show the surgical saw blade cartridge including an exemplary arrangement of reference features **56**, **56**A, **56**B, **56**C, **56**D. The plurality of reference features **56**, **56**A, **56**B, **56**C, **56**D may be located on the third plate **16**, the first and/or second plates **12**, **14** may be shaped to expose the reference features **56**, **56**A, **56**B, **56**C, **56**D. For example, as illustrated in FIG. **33**, the first and/or second plates **12**, **14** may comprise an opening **30**A, **94**A to provide access to the reference feature **56**. It is also contemplated that the reference feature(s) **56**A, **56**B, **56**C, **56**D may be positioned on a portion of the third plate **16** that is exposed from the first and/or second plates **12**, **14**. Referring to FIG. **33**, a plurality of reference features **56**A, **56**B, **56**C, **56**D are defined on the proximal portion **18**A of the third plate **16** that extends beyond the outer edge of the first and/or second plates **12**, **14**. While not illustrated, it is further contemplated that the first and/or second plates **12**, **14**, may comprise a cut-out, define a recess in an outer edge, and/or define additional apertured and/or openings to provide access to a the reference feature(s) **56 56**A, **56**B, **56**C, **56**D defined on the third plate **16**. As described above, the reference features **56**, **56**A, **56**B, **56**C, **56**D may be formed as an aperture, divot, engraving, color marking, or similar marking. Furthermore, while the reference features **56**, **56**A, **56**B, **56**C, **56**D are illustrated in the figures as being disposed on the third plate **16**, it is also contemplated that the reference features **56**, **56**A, **56**B, **56**C, **56**D may be disposed on the first and/or second plate(s) **12**, **14**. The reference features **56**, **56**A, **56**B, **56**C, **56**D for a given saw blade cartridge may be placed on any combination of the first, second, and/or third plate(s) **12**, **14**, **16**.

[0156] Referring to FIGS. **35**A and **35**B, an alternative configuration for a saw blade **70** any of the saw blade cartridges **10**, **110**, **210** described above is illustrated. The saw blade **70** includes a blade head **74** with a plurality of teeth **76**, **78**. The blade head **74** may further be formed with one or more apertures **75**. Referring to FIG. **35**A, an exemplary blade head **74** including a pair of apertures **75**A, **75**B is illustrated. Alternatively, FIG. **35**B illustrates an exemplary blade head **74** including a three apertures **75**A, **75**B, **75**C. It is contemplated that the blade head **74** may be formed with any number of apertures **75**. The apertures **75** may be configured to allow for removal of debris while cutting. It is also contemplated that the apertures **75** may be configured to provide cooling of the blade **70** and/or blade head **74** while cutting. Furthermore, the apertures **75** may be configured to assist in manufacturing the blade **70**. For example, when manufacturing the blade **70**, the apertures **75** may be machined and/or formed in the blank piece of material from which the blade will be formed. The apertures may then be used during manufacturing as a reference point, such that the blade body **72**, blade head **74**, and/or teeth **76**, **78** may be machined/formed suing the aperture(s) **75** as a reference. For example, the manufacturing process may be setup such that the apertures **75** are identifiable, and the machining tool is programmed to cut and/or grind the blank piece of the material at a location based on the aperture(s) **75** to form the teeth.

[0157] Referring to FIGS. **37** to **40**, an alternative configuration for an opening **30**, **94** in the guide bar **11** of any of the saw blade cartridges **10**, **110**, **210** described above is illustrated. As described above, either of the first and/or second plates **12**, **14** may define an opening **30**, **94** to provide access to a reference feature **56** that may be used for identifying and or registering the saw blade cartridges **10**, **110**, **210** to a navigation system. FIGS. **37** and **38** illustrate a guide bar **11** that may include alternative configuration of an opening **30**B, **94**B in the first and/or second plate **12**, **14**. The opening **30**B, **94**B may be shaped to define a first region **31**B, **93**B and a second region **33**B, **95**B. The first region **31**B, **93**B of the opening **30**B, **94**B may have a first size and be configured to receive a registration tool **2**. The second region **33**B, **95**B of the opening **30**B, **94**B may have a second size that is smaller than the size of the first region **31**B, **93**B. The second region **33**B, **95**B may further be configured to identify and/or guide the registration tool **2** to a contact point **35**B, **97**B. The contact point **35**B, **97**B may be located on the first and/or second plate(s) **12**, **14**. It is also contemplated that the second region **33**B, **95**B may further be configured to guide the registration tool **2** to a contact point or reference feature on the third plate **16**. For example, in operation, the registration tool **2** may be inserted into first region **31**B, **93**B of the opening **30**B, **94**B defined by the first and/or second plate **12**, **14**. The registration tool may be maneuvered within the opening **30**B, **94**B from the first region **31**B, **93**B to the second region **33**B, **95**B. The second region **33**B, **95**B may be sized and/or shaped such that the as the registration tool **2** is mover within the opening **30**B, **94**B, the second region **33**B, **95**B guides the registration tool **2** to contact point **35**B, **97**B that may correspond to a reference feature of the saw blade cartridges **10**, **110**, **210** that is recognizable and/or known by the navigation system. In this configuration of the guide bar **11**, it is contemplated that the aperture configuration of the reference feature **56** may be removed from the third plate **16**, and as the registration tool **2** is maneuvered within the opening **30**B, **94**B, the registration tool **2** may be in contact with the surface of the third plate **16**. As illustrated in FIGS. **37** and **38**, an exemplary configuration of the opening **30**B, **94**B may comprise a key-shape.

[0158] Referring a FIGS. **39** and **40**, another alternative configuration for an opening **30**, **94** in the guide bar **11** of any of the saw blade cartridges **10**, **110**, **210** described above is illustrated. Similar to the opening(s) **30**B, **94**B of FIGS. **37** and **38**, the configuration of the opening(s) **30**C, **94**C illustrated in FIGS. **39** and **40** may be shaped to define a first region **31**C, **93**C and a second region **33**C, **95**C. The first region **31**C, **93**C of the opening **30**C, **94**C may have a first size and be configured to receive a registration tool **2**. The second region **33**C, **95**C of the opening **30**C, **94**C may have a second size that is smaller than the size of the first region **31**C, **93**C. The second region **33**C, **95**C may further be configured to identify and/or guide the registration tool **2** to a contact point **35**C, **97**C. The contact point **35**C, **97**C may be located on the first and/or second plate(s) **12**, **14**. It is also contemplated that the second region **33**C, **95**C may further be configured to guide the registration tool **2** to a contact point or reference feature on the third plate **16**. As illustrated in FIGS. **39** and **40**, an exemplary configuration of the opening **30**B, **94**B may comprise a triangle-shape.

[0159] During the course of advancing the saw blade cartridge **10**, **110**, **210**, the saw blade cartridge **10**, **110**, **210** is exposed to resistive forces. The inner plate **16**, **116**, **216** provides structural strength to the guide bar **11**, **111**, **211**. This structural strength resists the extent to which the bar **11**, **111**, **211** would otherwise flex when exposed to these resistive forces. Further reinforcement of the guide bar **11**, **111**, **211** is provided by the sections of the outer tines **46**, **146**, **246** that extend forward of the head **58**, **158**, **258** of the inner tine **48**, **148**, **248**. The lobes **60**, **160**, **260** and/or tabs **261** provide structural strength to the guide bar **11**, **111**, **211** at the distal portion **20**, **120**, **220** of the guide bar **11**, **111**, **211**. This is the portion of the guide bar **11**, **111**, **211** initially exposed to the resistive forces. The presence of lobes **60**, **160**, **260** and/or tabs **261** thus further reduces the likelihood that the distal portion **20**, **120**, **220** of the guide bar **11**, **111**, **211** will skive or dive. The prevention of the skiving or diving of the distal portion **20**, **120**, **220** of the guide bar **11**, **111**, **211** reduces the likelihood that the saw blade cartridge **10**, **110**, **210** as whole will skive or dive.

[0160] A further feature of the saw blade cartridge **10**, **110**, **210** is that the drive links **62**, **162**, **262** are, on both sides, encased in the inner plate **16**, **116**, **216**. This means that, if a surgeon holds or touches the sides of the guide bar **11**, **111**, **211** when the saw blade cartridge **10**, **110**, **210** is actuated, the surgeon's fingers do not come into contact with the reciprocating driving links **62**, **162**, **262**. At a minimum, this makes it essentially impossible for the drive links **62**, **162**, **262** to, when moving, tear the glove off the surgeon's fingers. Still another benefit of this construction, is that it facilitates the efficient welding of the side portions of plates **12**, **14**, **16**, **112**, **114**, **116**, **212**, **214**, **216**. This side welding of the plates **12**, **14**, **16**, **112**, **114**, **116**, **212**, **214**, **216** so as to form side welds along the opposed sides of the guide bar **11**, **111**, **211** serves to strengthen the guide bar.

[0161] For example, the various features of the different configurations of the saw blade cartridge **10**, **110**, **210** can be combined in any arrangement of the various components.

[0162] Likewise, not all configurations of the saw blade cartridge **10**, **110**, **210** may have all the described features. For example, it is not required that all configurations of the saw blade cartridge **10**, **110**, **210** include outer tines that extend forward of the inner tine. In yet another example, it may not be necessary that all outer tines include inwardly directed lobes.

[0163] Likewise, in some versions of the saw blade cartridge **10**, **110**, **210**, the inner plate **16**, **116**, **216** may be wholly or partially formed with one or both of the first plate **12**, **112**, **212** or the second plate **14**, **114**, **214**. Thus, in these versions of the saw blade cartridge **10**, **110**, **210**, once a blank is formed, the blank is machined to define the slots in which the drive rods are seated and that define the perimeter of the tines. Alternatively, the opening-and slot-defining portion of the guide bar may be formed by molding.

[0164] In some versions of the saw blade cartridge **10**, **110**, **210**, a single drive link may be all that is needed to pivot the blade head. In some versions of the saw blade cartridge **10**, **110**, **210** the one or more drive links may not be parallel to the longitudinal axis through the cartridge. Thus, in these versions of the saw blade cartridge **10**, **110**, **210**, the openings in the guide bar through in which elements of the one or more drive links are seated likewise may not be aligned on line that are parallel to the longitudinal axis through the cartridge.

[0165] In versions of the saw blade cartridge **10**, **110**, **210** in which the at least one drive link and the blade are provided with complementary teeth, the guide bar may have a structure different from what has been described. Specifically, the guide bar may be formed so that head are which the blade head is not an integral part of one of the plates forming the bar. In these versions of the saw blade cartridge **10**, **110**, **210**, the structure around which the blade pivots may be a pin mounted to or integral with at least one of the plates that form the bar.

[0166] In some versions of the saw blade cartridge **10**, **110**, **210** two or all three of the plates that form the guide bar may not be three distinct plates that are welded or otherwise secured together. In these alternative versions of the saw blade cartridge **10**, **110**, **210**, the plural plates are machined or molded together as a single unit.

[0167] The various shapes of the elements may also vary from what has been described.

[0168] Accordingly, it is an object of the appended claims to cover all such variations and modifications that cover the true spirit and scope of the saw blade cartridge **10**, **110**, **210**.

Claims

1. A saw blade cartridge for use with a surgical device, the saw blade cartridge comprising: a guide bar including a distal portion and a proximal portion, the guide bar comprising: a first plate; a second plate; and an inner plate disposed partially between the first and the second plates, the inner plate defining a pivot surface; a blade at least partially disposed between the first and the second plates, the blade comprising: a blade body including a proximal end and a distal end; and a blade head disposed on the distal end of the blade body, the blade head formed with teeth, wherein the proximal end of the blade body abuts the pivot surface; wherein the proximal portion defines a proximal end; wherein one of the first and second plates define a first outer edge, wherein the inner plate defines a second outer edge, the second outer edge extends beyond the first outer edge defined by one of the first and second plates and wherein the second outer edge tapers outwardly from the proximal end to the distal end of the proximal portion such that the second outer edge assists in aligning the saw blade cartridge with the surgical device.
2. The saw blade cartridge of claim 1, wherein the guide bar further defines a longitudinal axis extending between the distal portion and the proximal portion of the guide bar; and wherein the second outer edge defines a first taper angle relative to the longitudinal axis between a first point and a second point on the second outer edge; and wherein the second outer edge defines a second taper angle relative to the longitudinal axis between the second point and a third point on the second outer edge, the first point being closer to a proximal end of the cartridge than the third point and the second point being disposed between the first point and the third point.
3. (canceled)
4. The saw blade cartridge of claim 1, wherein the inner plate further comprises a tine extending from a proximal end of the guide bar, the tine defining the pivot surface.
5. (canceled)
6. The saw blade cartridge of claim 4, wherein the tine is an inner tine, and further comprising a first outer tine and a second outer tine, the first and second outer tines being positioned between the first and second plates and the inner tine being disposed between the first and second outer tines.
7. The saw blade cartridge of claim 6, wherein the first and second outer tine tines extend closer to the distal end of the guide bar than the inner tine.
8. The saw blade cartridge of claim 6, wherein the first and second outer tines each comprises a distal end; wherein the guide bar is configured such that the distal end of each of the first and second outer tines is proximal to a distal end of the first and the second plates; and wherein a side defined by the first and the second plates defines an aperture proximal to the distal end of the first and the second plates and distal to the distal end of each of the first and second outer tines, the aperture provides for debris exit.
9. (canceled)
10. The saw blade cartridge of claim 1, wherein the pivot surface is an arcuate surface; and wherein the proximal end of the blade body defining a recess that abuts the pivot surface, such that the blade body is configured to pivot about the pivot surface defined by the inner plate when the blade body is actuated.
11. (canceled)
12. The saw blade cartridge of claim 10, wherein the teeth define a thickness of the blade head that is equal to or greater than a thickness of the guide bar to allow the insertion of the guide bar into a kerf cut by the blade head.
13.-**15**. (canceled)
16. The saw blade cartridge of claim 1, wherein the inner plate extends laterally beyond the outer edge of the first and second plates.
17.-**18**. (canceled)

19. The saw blade cartridge of claim 1, wherein at least one of the plates defines a reference feature configured to facilitate determination of a position and/or orientation of the surgical blade with a navigation system.
20. The saw blade cartridge of claim 1, wherein the inner plate further comprises a reference feature configured to facilitate determination of a position and/or orientation of the surgical blade with a navigation system.
21. The saw blade cartridge of claim 19, wherein the reference feature is selected from the group comprising an optical marking, a divot, or an aperture.
22. The saw blade cartridge of claim 19, wherein the reference feature comprises a plurality of laser markings; and wherein at least one of the plurality of laser markings is positioned on each side of the longitudinal axis.
23.-**24**. (canceled)
25. The saw blade cartridge of claim 19, wherein the reference feature is configured to identify the saw blade cartridge to the navigation system, including one or more characteristics of the saw blade cartridge; and wherein the characteristic including at least one of the following: a type of the blade, a blade thickness of the blade head, a tooth configuration of the blade head, a length of the guide bar, and/or a width of the guide bar.
26.-**107**. (canceled)
108. A saw blade cartridge for use with a surgical device, the saw blade cartridge comprising: a guide bar including a distal portion and a proximal portion, the guide bar comprising: a first plate; a second plate; the first plate coupled to the second plate and the first plate including a pivot surface; a blade comprising: a blade body including a proximal end and a distal end; and a blade head disposed on the distal end of the blade body and configured to extend from the distal portion of the guide bar, the blade head formed with teeth, wherein the proximal end of the blade body abuts the pivot surface; wherein the proximal portion defines a proximal end; wherein the first plate defines a first outer edge; wherein the second plate defines a second outer edge; the guide bar is configured such that the first outer edge extends beyond the second outer edge; and wherein the first outer edge tapers outwardly from the proximal end to the distal end of the proximal portion of the guide bar such that the first outer edge assists in aligning the saw blade cartridge with the surgical device.
109.-**116**. (canceled)
117. A surgical saw system, the system comprising: a surgical device including a mount, the mount defining a mount surface, and the mount comprising: a first wall and a second wall extending above the mounting surface, the first and second wall positioned on opposing sides of a longitudinal axis of the mount; at least one tab disposed on each of the first and second walls, each of the at least one tabs positioned above and extending over the mounting surface; a saw blade cartridge removably couplable to the mount of the surgical device, the saw blade cartridge comprising: a guide bar including a distal portion and a proximal portion, the guide bar comprising: a first plate; a second plate; and an inner plate disposed partially between the first and the second plates, wherein one of the first and second plates define a first outer edge, wherein the inner plate is configured to define a second outer edge, the second outer edge extends beyond the first outer edge defined by one of the first and second plates; wherein the first and second walls are tapered relative to the longitudinal axis of the mount to facilitate alignment of the saw blade cartridge along the longitudinal axis; and wherein each of the first and second walls engage the second outer edge of the inner plate and an under surface of each of the at least one tabs engage a top surface of the inner plate.
118.-**128**. (canceled)
129. The saw blade cartridge of claim 108, wherein at least one of the plates defines a reference feature configured to facilitate determination of a position and/or orientation of the surgical blade with a navigation system; and wherein the reference feature is selected from the group comprising an optical marking, a divot, or an aperture.

130. The saw blade cartridge of claim 129, wherein the reference feature is configured to identify the saw blade cartridge to the navigation system, including one or more characteristics of the saw blade cartridge; and wherein the characteristics include at least one of the following: a type of the blade, a blade thickness of the blade head, a tooth configuration of the blade head, a length of the guide bar, and/or a width of the guide bar.
131. The surgical saw system of claim 117, wherein at least one of the plates defines a reference feature configured to facilitate determination of a position and/or orientation of the surgical blade with a navigation system; and wherein the reference feature is selected from the group comprising an optical marking, a divot, or an aperture.
132. The surgical saw system of claim 131, wherein the reference feature is configured to identify the saw blade cartridge to the navigation system, including one or more characteristics of the saw blade cartridge; and wherein the characteristics include at least one of the following: a type of blade, a blade thickness of a blade head, a tooth configuration of the blade head, a length of the guide bar, and/or a width of the guide bar.